Smart Monitoring Solutions for Real-Time Water pH Regulation in Aquatic Ecotoxicology

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Abstract

This study designs the mixed Exponentially Weighted Moving Average – Cumulative Sum (MEC) control chart, a statistical process control tool that effectively detects small and moderate shifts in process parameters, addressing a critical need in quality monitoring. The MEC chart leverages the strengths of both Exponentially Weighted Moving Average (EWMA) and Cumulative Sum (CUSUM) methodologies to enhance sensitivity while reducing false alarms, thus improving detection performance in various applications. This methodology is applied in a real-life context within an aquatic ecotoxicology laboratory, where daily monitoring of water pH levels is essential for safeguarding the health of sensitive aquatic organisms, such as mysids. The laboratory environment is meticulously controlled to simulate natural conditions, and our application of the MEC chart ensures that any deviations from the optimal pH level are detected promptly, thereby maintaining water quality and supporting the reliability of experimental outcomes.

The paper comprehensively evaluates the performance of the MEC chart in both zero-state and steady-state conditions, offering valuable insights for practitioners in the field. We present empirical evidence demonstrating that the MEC chart significantly outperforms traditional control charts, including Shewhart, CUSUM and EWMA, particularly in detecting small to moderate shifts in water pH levels. Furthermore, we provide optimal parameter settings tailored for specific monitoring scenarios, enhancing the applicability of proposed chart for quality control in laboratory environments.

Keywords: Aquatic Ecotoxicology, Conditional Expected Delay, Control Charts, Real-Time Monitoring, Water pH Regulation.

1. Introduction

Monitoring water quality is critical in ecotoxicology laboratories, where researchers study the impact of various chemicals on aquatic life. Aquatic ecosystems are particularly sensitive to changes in water chemistry, with potential Hydrogen (pH) levels playing a vital role in maintaining the health and viability of the organisms within these environments. The pH level indicates the acidity or alkalinity of water, which can significantly influence biological processes, including enzymatic activities, nutrient availability, and overall organismal health. As highlighted by [1], understanding the impact of physicochemical parameters, including pH, is fundamental to evaluating aquatic ecosystems. Fluctuations in pH can lead to physiological stress in aquatic organisms, resulting in altered behavior, impaired reproduction, and even increased mortality rates. In particular, indicator species like mysids are widely used in toxicity assessments due to their ecological relevance and sensitivity to changes in environmental conditions. Therefore, maintaining optimal pH levels is not only essential for the welfare of these organisms but also for the integrity of the research conducted in these laboratories. Given the importance of pH in experimental outcomes, as emphasized by [2], ecotoxicology labs must implement effective strategies for continuous monitoring and regulation of this critical parameter. However, maintaining a stable pH level can be challenging due to various external and internal factors, such as the introduction of chemicals, biological activity, and fluctuations in environmental conditions (see [3]). Even minor deviations from the acceptable pH range can result in significant consequences for both the aquatic organisms under study and the validity of the experimental results. As such, it is crucial for laboratories to adopt robust monitoring protocols to detect any variations promptly. This ensures not only the health of the aquatic life being studied but also the reliability of the data generated, which can influence regulatory decisions and environmental policies.

To achieve these monitoring objectives, laboratories can implement Statistical Process Control (SPC) techniques, which provide systematic methods for monitoring and improving process quality. SPC includes various tools and methodologies that allow for the identification and analysis of variations in processes, facilitating timely interventions to maintain optimal conditions, as outlined by [4]. Among these tools, control charts are instrumental in visualizing process behavior over time, allowing practitioners to detect trends and shifts in critical parameters like pH levels. It has found a wide range of applications in many areas, including monitoring processes in physical and pharmaceutical sciences; for example, see [5], [6], [7] and references therein. By employing control charts, laboratories can gain real-time insights into their processes, ensuring they remain stable and within established control limits.

Control charts are graphical tools used in SPC to monitor process parameters over time, distinguishing between common cause variation (natural fluctuations inherent to the process) and special cause variation (variations resulting from identifiable factors). The fundamental structure of a control chart consists of a central line, representing the process mean, and control limits that define the acceptable range of variation. Data points plotted above or

below these limits indicate that the process may be out of control, signaling the need for further investigation. An in-control (IC) situation occurs when all data points fall within the control limits, suggesting that the process variation is due solely to common causes and is thus stable. Conversely, an out-of-control (OOC) situation is identified when one or more data points fall outside the control limits or when a run of points shows a consistent trend, indicating the presence of special causes that may require corrective action. While Shewhart-type control charts by [8] are widely used due to their simplicity and ease of interpretation, they exhibit significant deficiencies in detecting small and moderate shifts in process parameters. Shewhart charts rely only on current sample for decision-making, making them less sensitive to subtle changes in the process mean. This insensitivity can result in delayed detection of emerging issues, potentially leading to substantial negative impacts on product quality and safety. Consequently, alternative charting methods have been developed to address these limitations. For instance, the Exponentially Weighted Moving Average (EWMA) control chart, which was introduced by [9], improves sensitivity by placing greater emphasis on recent data. Similarly, the Cumulative Sum (CUSUM) control chart, developed by [10], effectively detects shifts in process parameters by accumulating deviations over time. These memory-type charts utilize historical data to provide enhanced sensitivity to small shifts, allowing for more timely detection of changes in critical process parameters like pH levels in aquatic ecotoxicology laboratories.

The development of the CUSUM and EWMA charts marked significant progress in process monitoring by addressing the limitations of traditional Shewhart charts. However, researchers have continued to build upon these foundational techniques, seeking to enhance their sensitivity to small and moderate shifts in process parameters. This ongoing effort has resulted in a variety of modifications and improvements in both the CUSUM and EWMA control charts. Since their inception, many modifications of the EWMA and CUSUM charts have been explored, with researchers striving to improve their sensitivity and adaptability in different process monitoring scenarios. [11] introduced the mixed EWMA-CUSUM (MEC) control chart, combining the EWMA statistic as an input in the CUSUM chart for monitoring process location shifts. This was the first formal attempt to integrate two classical charts, offering improved sensitivity in detecting small shifts. They compared the performance of MEC chart to that of classical schemes, concluding that the MEC chart generally outperformed both the EWMA and CUSUM charts for small and moderate shifts.

Following [11], [12] proposed further enhancements to the MEC chart by incorporating head starts and run rules features. These modifications aimed to optimize the ability of MEC to quickly detect shifts, especially in processes where small shifts occur early, during the start-up phase. Their findings revealed improved performance in detecting these small shifts, a crucial consideration for early process monitoring. [13] carried out a comprehensive comparative study of the MEC chart alongside other memory-type control charts, such as EWMA and CUSUM, under various scenarios, including unknown process parameters. Their research evaluated different location estimators and used various run length distribution properties to measure performance. While no single estimator or chart proved superior

across all situations, the MEC chart showed robustness in contaminated environments, and the EWMA chart using the median estimator had the best average performance.

[14] extended the application of MEC chart to multivariate processes, reflecting the growing need for methods that handle the complexities of monitoring multiple variables simultaneously. This multivariate adaptation widened the utility of the MEC chart beyond univariate monitoring, allowing for more comprehensive process control. [15] contributed another significant improvement by introducing a median-based version of the MEC chart. Their research addressed the challenge of outliers in contaminated environments, proposing the median MEC as a robust alternative for environments prone to data contamination. In the same year, [16] developed a mixed chart for processes following the Conway Maxwell Poisson distribution, extending the mixed chart approach into count data monitoring. Their study showed superior performance of the mixed chart in detecting moderate shifts, addressing a niche in process control for count data applications. [17] incorporated auxiliary variables into a combined MEC chart for simultaneous monitoring of both location and dispersion, providing a dual-purpose chart capable of handling multiple process characteristics in real-time monitoring. In a related study, [18] developed an effective statistical process control scheme tailored for monitoring industrial effluents, specifically targeting non-normal distributions of zinc in wastewater from manufacturing industries. Their optimal scheme minimizes total costs, including quality and manpower, while ensuring compliance with allowable false alarm rates.

Another recent developments include the work by [19], who introduced the mixed modified EWMA-EWMA (MEME) chart. This new chart structure was designed for improved detection of small to moderate shifts in the process location, demonstrating the ongoing interest in creating hybrid charts that can enhance sensitivity across various shift magnitudes. Lastly, [20] provided a thorough evaluation of various extensions of the EWMA chart, including Double, Triple, and Quadruple EWMA charts. Their study concluded that these new chart structures outperformed the classical EWMA chart, particularly in both zero- and steady-state conditions. Their findings highlight the continuous evolution of control charts and the drive to improve process monitoring techniques for enhanced sensitivity and robustness.

Also drawing inspiration from the MEC chart, [21] introduced a reverse version called the mixed CUSUM-EWMA (MCE) chart for monitoring process location, where the CUSUM statistic is used as an input in the EWMA structure. This reverse configuration was further extended by [22] for dispersion monitoring, resulting in the MCE chart for variance shifts. Extensive comparisons with existing methods demonstrated that the MCE chart, in both its location and dispersion forms, was generally more effective than other competing charts under various process conditions, making it another significant advancement in the field of control charts.

Building upon these developments, the current study focuses on several key contributions to further the application and improvement of control charts. First, the paper moves beyond the traditional focus on zero-state Average Run Length (ARL) metrics, which measure performance assuming the process starts in a stable condition. Instead, we also consider steady-state properties to provide a more comprehensive evaluation of chart performance.

This approach contrasts with many previous studies that have emphasized zero-state metrics alone. Second, this research offers guidance for quality practitioners by identifying optimal parameter settings for the MEC chart, depending on the magnitude of process shifts that need to be detected. By providing this practical insight, the study ensures that practitioners can make informed decisions on chart configuration. Finally, the study applies the MEC chart to real-world data, specifically monitoring water pH levels in an ecotoxicology laboratory. This application demonstrates the utility of MEC chart in real-time monitoring.

The rest of the article is organized as follows: Section 2 discusses the basic structures of classical control charts and the MEC chart, along with the performance measures used to evaluate chart efficiency. Section 3 presents the numerical results and detailed discussions. In Section 4, comparisons between the MEC, EWMA, and CUSUM charts are provided. Section 5 demonstrates the application of these control charts in monitoring water pH levels, while Section 6 concludes the article and provides recommendations for future use of the MEC chart in process monitoring.

2. Methodology

In this section, we outline the fundamental structures of several classical control charts along with the designed MEC control chart. To begin, we assume that the quality characteristic of interest, denoted as X_{ij}, is normally distributed with a mean μ and a known, stable variance σ_0^2, i.e., X_{ij} follows a normal distribution $N(\mu, \sigma_0^2)$, for all i = 1, 2, … and j = 1, 2, …, n, where i represents the sample number and j denotes the observation within each sample. The mean is expressed as $\mu = \mu_0 + \delta\frac{\sigma_0}{\sqrt{n}}$, where μ_0 is the IC process mean, n is the sample size, and δ represents the magnitude of the process shift, measured in units of $\frac{\sigma_0}{\sqrt{n}}$. This section assumes the process variance is constant and known, allowing for the sample mean for the ith sample to be computed as follows:

$$\bar{X}_i = \frac{\sum_{j=1}^{n} X_{ij}}{n}. \tag{1}$$

Following this, we discuss the Shewhart, EWMA, and CUSUM control charts in detail, each offering distinct methods for detecting shifts in the process mean, along with their respective control structures. Finally, the MEC chart, which integrates features from both EWMA and CUSUM charts, is introduced, along with various performance measures used to evaluate the efficiency of these charts under different conditions.

2.1. The Classical Shewhart $\overline{X}$ Chart

The Shewhart $\bar{X}$-chart, developed by [8], is a widely used control chart for detecting significant changes in the process mean. It is a memoryless chart, meaning it only considers information from the current sample without incorporating past data. The statistic plotted on the Shewhart chart is the sample mean, $\bar{X}_i$, which is computed as shown in equation (1). This

statistic is plotted against the lower and upper control limits (LCL_S and UCL_S), defined as follows:

$$\left.\begin{aligned} LCL_s &= \mu_0 - L_s \frac{\sigma_0}{\sqrt{n}} \\ UCL_s &= \mu_0 + L_s \frac{\sigma_0}{\sqrt{n}}. \end{aligned}\right\} \tag{2}$$

Here, L_S is the control limit coefficient that determines the width of the control limits. A common choice for L_S is 3, which results in the well-known 3-sigma limits. If the sample mean $\bar{X}_i$ falls outside these control limits, it signals that the process may be OOC, meaning a significant shift in the process mean has occurred.

Despite its simplicity and ease of use, the Shewhart chart is not very effective for detecting small to moderate shifts in the process mean. Its primary strength lies in identifying large shifts quickly. For minor shifts, the chart may not signal promptly, leading to potential delays in corrective actions. Therefore, alternative control charts such as the EWMA and CUSUM charts are often preferred for processes where smaller shifts are of concern.

2.2. The CUSUM Control Chart

The CUSUM control chart is a memory-type chart designed to detect small and moderate shifts in the process mean more effectively than the Shewhart chart. Unlike the Shewhart chart, the CUSUM chart incorporates information from all past samples, making it more sensitive to gradual changes in the process. The CUSUM chart can be implemented in two forms: the tabular CUSUM and the V-mask CUSUM. The tabular CUSUM is more commonly used in practice, where separate one-sided CUSUMs are calculated for detecting upward and downward shifts. The upward and downward CUSUMs are defined as:

$$\left.\begin{aligned} C_i^- &= \max[0, C_{i-1}^- + (\mu_0 - K_C) - \bar{X}_i] \\ C_i^+ &= \max[0, C_{i-1}^+ + \bar{X}_i - (\mu_0 + K_C)], \end{aligned}\right\} \tag{4}$$

where $K_C = k_C \frac{\sigma_0}{\sqrt{n}}$ is the sensitivity parameter of CUSUM chart. If either C_i^- or C_i^+ exceeds the decision interval $H_C = h_C \frac{\sigma_0}{\sqrt{n}}$, it indicates that the process mean has shifted. The strength of the CUSUM chart lies in its ability to detect small, sustained shifts in the process mean. As it accumulates information over time, it quickly picks up on trends that may go unnoticed by the Shewhart chart. However, its sensitivity to small shifts comes with the trade-off of being slower in detecting large shifts compared to the Shewhart chart. As Albert Einstein once said, "Not everything that can be counted counts, and not everything that counts can be counted." The CUSUM chart exemplifies this, as it captures the cumulative effect of small changes that individually may not seem significant, but collectively indicate a problem in the process.

2.3. The Classical EWMA Control Chart

The EWMA control chart also addresses some limitations of the Shewhart chart by placing more weight on recent data points while still incorporating historical information. This

characteristic makes the EWMA chart particularly effective for detecting small shifts in the process mean. The EWMA statistic is calculated using the formula:

$$EWMA_i = \lambda_E \bar{X}_i + (1 - \lambda_E) EWMA_{i-1}. \tag{5}$$

In this equation, $EWMA_i$ represents the current EWMA statistic and λ_E is the sensitivity parameter that determines the weight assigned to the most recent sample. The value of λ_E lies between 0 and 1, with larger values placing more emphasis on recent data. When λ_E equals 1, the EWMA chart becomes equivalent to the Shewhart chart.

The control limits for the EWMA chart are established based on the expected value and asymptotic variance of the EWMA statistic. The limits are given by:

$$\left.\begin{aligned} LCL_E &= \mu_0 - L_E \frac{\sigma_0}{\sqrt{n}} \sqrt{\frac{\lambda_E}{(2 - \lambda_E)}} \\ UCL_E &= \mu_0 + L_E \frac{\sigma_0}{\sqrt{n}} \sqrt{\frac{\lambda_E}{(2 - \lambda_E)}}. \end{aligned}\right\} \tag{6}$$

Here, L_E determines the width of the control limits. A well-chosen L_E value balances the sensitivity and specificity of the chart, minimizing false alarms while effectively detecting shifts.

One of the primary advantages of the EWMA chart is its ability to quickly signal small shifts in the process mean. By continuously updating the statistic with each new sample, the EWMA chart is more responsive to changes compared to the Shewhart chart. This responsiveness is particularly valuable in industries where maintaining a consistent process is crucial, as it allows for timely interventions before significant deviations occur (see [23]). In practice, practitioners can optimize the parameters of the EWMA chart, such as λ_E and L_E, based on the specific characteristics of the process and the shifts of interest. [24] provides guidance on selecting optimal values for these parameters, emphasizing that lower λ_E values are preferable for detecting minor deviations, while higher values are better suited for larger shifts.

2.4. The Mixed EWMA-CUSUM Control Chart

The MEC control chart combines the strengths of both EWMA and CUSUM control charts to enhance the detection of small shifts in the process mean. Instead of relying solely on raw observations, the MEC chart utilizes the EWMA statistic as input into the CUSUM framework, thus allowing for a more responsive monitoring system. The MEC chart calculates two monitoring statistics: the upper MEC statistic (MEC_i^+) and the lower MEC statistic (MEC_i^-). These statistics are defined as follows:

$$\left.\begin{aligned} MEC_i^- &= \max[0, MEC_{i-1}^- + (\mu_0 - K_M) - EWMA_i] \\ MEC_i^+ &= \max[0, MEC_{i-1}^+ + EWMA_i - (\mu_0 + K_M)]. \end{aligned}\right\} \tag{7}$$

In this equation, $EWMA_i$ is derived from the current sample mean using the EWMA formula, and K_M is a reference parameter that is set to a predetermined value based on the expected shift. The initial values for both MEC statistics are zero (i.e., $MEC_0^- = MEC_0^+ = 0$).

To assess whether the process is OOC, the MEC statistics are compared against a control limit (H_M). An upward shift in the process average is indicated when MEC_i^+ exceeds H_M, while a downward deviation is signaled if MEC_i^- exceeds H_M. The design parameters of the MEC chart include the sensitivity parameter from the EWMA chart (denoted by λ_M for MEC chart) and the reference values K_M and H_M. These parameters are determined in a manner analogous to the CUSUM chart, where K_M and H_M are expressed as follows:

$$\left.\begin{aligned} K_M &= k_M \frac{\sigma_0}{\sqrt{n}} \sqrt{\frac{\lambda_M}{2 - \lambda_E}} \\ H_M &= h_M \frac{\sigma_0}{\sqrt{n}} \sqrt{\frac{\lambda_M}{2 - \lambda_E}}. \end{aligned}\right\} \tag{8}$$

Here, k_M and h_M serve similar purposes as in the traditional CUSUM scheme. The value of h_M is identified by fixing λ_M and k_M, and searching for the corresponding h_M that achieves the desired IC average run length (ARL_0). The unique structure of MEC chart allows it to efficiently detect both small and moderate shifts in process mean, making it a valuable tool in the field of SPC. The EWMA control chart exemplifies the principle that informed decision-making relies on understanding both current and historical data, enabling effective process monitoring and improvement.

2.5. Performance Measures

Performance measures are crucial metrics that allow for the assessment of the efficiency of control charts and enable comparisons between competing schemes. Historically, traditional hypothesis testing measures of type I and type II errors served as performance measures for control charts until [25] introduced the concept of ARL. The run length (RL) is defined as the number of units or samples inspected until a signal is generated, which may indicate either a true OOC condition or a false alarm. Two important metrics in this context are ARL_0, which represents the average run length when the process is IC, and ARL_1, which denotes the average run length when the process is OOC. Ideally, the goal is to maximize ARL_0 when the process is in a state of control and minimize ARL_1 when the process is OOC. The evaluation of performance typically relies on the characteristics of the RL distribution. The following outlines key performance measures utilized in the analysis of control charts:

Zero-state Average Run Length: The zero-state average run length is the expected value of the RL when the computation begins at the start of the monitoring period (i.e., at time $i = 1$). It is expressed as $\mathrm{ARL}_{\text{zero-state}} = \mathbb{E}(RL)$. This measure helps assess the performance of the chart under initial conditions.

Conditional Expected Delay: Conditional Expected Delay (CED) quantifies the average number of samples required to detect a change after it has occurred, given that the process was in control prior to the change and there were no previous false alarms. CED can be mathematically expressed as $CED_\tau = \mathbb{E}\left[(RL - \tau + 1)|_{(RL \geq \tau)}\right]$, where τ denotes the unknown point at which the change occurs. This measure is particularly useful for evaluating the responsiveness of the control chart, as demonstrated by [26] in their recent study comparing the performance of several control charts.

Steady-state Average Run Length: The steady-state average run length ($ARL_{\text{steady}-\text{state}}$) is similar to its zero-state counterpart, but the computation begins when the process has been operating for a while without prior false signals and is in a stable condition. It is defined as $\text{ARL}_{\text{steady-state}} = \lim_{\tau \to \infty} \mathbb{E}\left[(RL - \tau + 1)|_{(RL \geq \tau)}\right]$. The steady-state ARL represents the limit of CED as τ approaches infinity. Recent studies, such as [27], have integrated this metric into their evaluations to provide a more comprehensive performance assessment.

Extra Quadratic Loss: Extra Quadratic Loss (EQL) is an important performance measure that accounts for the uncertainty in the magnitude of shifts in process parameters. Given that the exact shift magnitude is rarely known in advance, practitioners often seek a chart that exhibits good overall performance across a range of shifts. EQL is defined as the weighted expected ARL that considers shifts from a minimum to a maximum range (i.e., $\delta_{\max} - \delta_{\min}$) and uses the square of the shift as the weight. Mathematically, it is represented as $EQL = \frac{1}{\delta_{max} - \delta_{min}} \int_{\delta_{min}}^{\delta_{max}} \delta^2 \, ARL(\delta) d\delta$, where $ARL(\delta)$ denotes the OOC ARL for a shift of size δ.

Absolute Expected Loss: Similar to EQL, Absolute Expected Loss (AEL) is another overall performance measure that provides an expected weighted ARL using the absolute value of the shift as the weight function. It is expressed mathematically as $AEL = \frac{1}{\delta_{max} - \delta_{min}} \int_{\delta_{min}}^{\delta_{max}} |\delta| \, ARL(\delta) d\delta$. This measure is particularly useful for evaluating the performance of control charts across a range of potential shifts. The choice of using the absolute value of delta instead of delta squared in the weight function is significant. In EQL, the weight is proportional to δ^2, which assigns disproportionately low weight to shifts smaller than 1, as the square of a value less than 1 approaches zero. Conversely, shifts larger than 1 receive amplified weight due to squaring. AEL addresses this limitation by using the absolute value of delta, ensuring that all shifts, regardless of their size, contribute appropriately to the overall performance assessment of the control chart.

Sequential Extra Quadratic Loss: Sequential EQL (SEQL) refers to an EQL calculated progressively over a series of shifts. The evaluation starts with the smallest shift and incorporates one additional shift at each step until the final shift is reached. The final SEQL value corresponds to the EQL for the entire range of shifts. This measure aids in assessing the overall performance of a control chart across specific shift ranges.

These performance measures provide a comprehensive framework for evaluating the effectiveness of these charts across various operational scenarios. In the subsequent section,

we will present the results of our comparative study, discussing the performance metrics of the MEC chart in detail and illustrating its advantages in monitoring process shifts.

3. Results and Discussion

In this section, we evaluate the performance of MEC control chart using key metrics such as the ARL, EQL, and AEL under both zero-state and steady-state conditions. Other important run length properties, including the median (MDRL), standard deviation (SDRL), and selected percentiles (P_5RL, P_{25}RL, P_{75}RL, and P_{95}RL), are also investigated for various values of the sensitivity parameters k_M and λ_M, as shown in supplementary Tables S1–S4. The analysis encompasses small, moderate, and large parameter values to study their effects on the detection performance of chart.

We explore a range of k_M values from 0.125 to 3 and λ_M values from 0.01 to 1. The special case of λ_M equal to 1 simplifies the MEC chart to the classical CUSUM chart, while $\lambda_M = 1$ and $k_M = h_m$ corresponds to the classical Shewhart chart. A Monte Carlo simulation approach is employed to estimate the performance of the MEC control chart by generating random samples and calculating the ARL for different shifts in the process mean. The simulation uses 10^6 iterations to compute the ARL, and a search is performed to determine the decision interval h_M that yields ARL_0 approximately equal to 370.

Given the high volume of simulation results produced from the sensitivity analysis, we report only the most relevant results, while supplementary tables are available in the appendix. Each table presents results for a specific fixed value of k-star, covering all combinations of λ_M and process mean shifts (δ), ranging from 0 to 6 in increments of 0.25.

The following algorithm outline the procedure used to compute the MEC ARL using simulation:

1. Specify τ (where $\tau = 1$ corresponds to the zero-state condition).
2. Generate random samples based on the following:
   - For sample number i less than τ, X_i is normally distributed with mean μ_0 and variance σ_0^2.
   - For sample number i greater than or equal to τ, X_i is normally distributed with mean μ and variance σ_0^2.
3. Compute the plotting statistics using the relevant formulas (equations 5 and 7).
4. Compare the plotting statistics with the control limit (from equation 8) and denote the first i for which the plotting statistics exceed the limit as r.
5. If r is less than τ, discard the value; otherwise, save it.
6. Repeat steps 2 to 5 a total of 10^6 times.
7. Compute the ARL by taking the mean of r, conditional on r being greater than or equal to τ, and subtract $\tau + 1$ from the result.

The performance of MEC control chart across various parameter settings is discussed in the following subsections.

3.1. Performance Evaluation Under Zero-State Condition

Supplementary tables S1 to S9 present the zero-state ARL profiles of the MEC chart with the ARL_0 set at 370. Each table provides values corresponding to different combinations of k_M and λ_M, covering small, moderate, and large values of δ. For a fixed value of k_M, it is observed that the standard deviation of run length $SDRL_0$ tends to increase with an increase in λ_M. However, for OOC situations, the SDRL fluctuates, generally rising with an increase in λ_M, except when k_M is large. According to [28], this behavior suggests that MEC charts with larger values of λ_M have a higher false alarm rate than those with smaller λ_M. We also evaluate the MDRL, a robust alternative to ARL, since run length distributions are often skewed, making the mean less representative. When k_M is small, the MDRL tends to decrease as λ_M increases. However, this trend changes for larger values of k_M (approximately 1.5 and up), where moderate values of λ_M yield smaller MDRLs than either very small or very large values of λ_M.

The run length distribution of the MEC chart is positively skewed, as is the case with most control charts. This skewness is more pronounced for IC run lengths and small shifts in the process mean. For example, in Table S1 where k_M is small (e.g., k_M = 0.125, λ_M = 0.01, and δ = 2.00), the P_{95}RL is 34, while it drops to 10 when λ_M is 0.75 for the same shift size. This indicates that the distribution of run lengths becomes tighter and less variable as λ_M increases. However, for large values of k_M and λ_M, the distribution shows significant variability across different parameter settings, especially in detecting small and moderate shifts.

As λ_M approaches 1, the MEC chart behaves more like the classical CUSUM chart, and its detection ability improves, particularly for large shifts. For small values of k_M in Table S1 (e.g., k_M = 0.125), the classical CUSUM chart outperforms the MEC chart, but this trend reverses as k_M increases. For instance, at k_M = 0.5 in Table S2, the smallest ARL is observed when λ_M is around 0.1 for small shifts (δ < 0.5), while larger values of λ_M are required to detect larger shifts more effectively.

For moderate values of k_M in Tables S5 and S6 (around 1–1.5), small values of λ_M (e.g., λ_M = 0.01) give better performance for detecting small shifts. For example, at k_M = 1, a shift of 0.25σ requires about 196 samples on average when λ_M = 1, compared to only 71 samples when λ_M = 0.01. This trend holds for shifts up to δ = 0.75, after which larger values of λ_M become more effective in detecting significant shifts.

For large values of k_M in Tables S7 – S9, it becomes difficult to achieve an ARL_0 of 370 with small values of λ_M, so only larger values of λ_M (close to 1) are considered in these cases. In the most extreme case, when k_M = 3, only λ_M = 1 is possible, corresponding to the classical Shewhart chart. Large values of k_M (≥ 2) perform best when combined with small values of λ_M (0.05–0.175) for detecting small to moderate shifts, while larger shifts require larger values of λ_M for quick detection.

Overall, no single combination of k_M and λ_M provides the best ARL performance across all shift sizes. The selection of parameters should therefore depend on the type of shift (small, moderate, or large) that needs to be detected quickly and the nature of the process being monitored.

Zero-State EQL Performance: The EQL metric assesses the overall performance of a control chart over the entire range of shifts. Smaller EQL values indicate better detection ability of shifts. There are certain parameter combinations for which the MEC chart with ARL_0 = 370 does not exist, as indicated by the (*) symbol in Tables 1 and 2.

[Please insert Table 1 here]

From Table 1, it is evident that some values of the parameters (k_M and λ_M) lead to extremely large EQL values, indicating poor overall performance. However, for a fixed k_M, the EQL drastically decreases as λ_M increases. For example, when k_M = 0.125, the EQL reduces by approximately 82% when λ_M changes from 0.01 to 1. The same trend can be observed when λ_M is fixed, and k_M is increased. For instance, fixing λ_M at 0.01 results in the EQL dropping from 267.91 at k_M = 0.125 to 61.00 at k_M = 1.5.

It is evident that the overall performance improves when both k_M and λ_M are set at larger values. The best EQL from Table 1 is 19.581 at k_M = 1 and λ_M = 1, which corresponds to the classical CUSUM chart. The best EQL for the MEC chart is 19.583 at k_M = 1.5 and λ_M = 0.75. Both values are nearly identical, indicating a close competition between the MEC and CUSUM charts.

Zero-State AEL Performance: The AEL, like EQL, measures the overall performance of control charts. However, because EQL gives too much weight to large shifts by squaring the shift magnitude, AEL provides a more balanced perspective by using the absolute value of the shift. The AEL results are given in Table 2.

[Please insert Table 2 here]

As expected, the AEL values are much smaller than their corresponding EQL values. For small values of k_M and λ_M, the AEL is quite large, indicating poor overall performance. In particular, the row for λ_M = 1 in Table 2 corresponds to the AEL values for the classical CUSUM chart, which shows the smallest values across most columns except for $k_M \geq 1$. This suggests that the MEC chart is inferior to the classical CUSUM chart when $k_M < 1$.

Nevertheless, the minimum AEL in Table 2 is 8.389, corresponding to the MEC chart with k_M = 2.5 and λ_M = 0.25. In general, moderate to large values of both k_M and λ_M produce smaller AEL values, indicating better overall performance in these regions.

3.2. Performance Evaluation Under Steady-State Condition

In this section, we reevaluate the MEC chart properties under steady-state conditions, focusing primarily on its ARL characteristics. Steady-state ARLs are calculated after the process has been running for some time and has stabilized, which allows for a fair comparison between different control chart designs.

For this analysis, the change point is set at τ = 201. Various other values of τ, ranging from τ = 51 to τ = 501, were considered. The ARL values of MEC chart appears to be most stable at τ = 501, but for computational efficiency, we have set τ = 201, as the results are nearly identical to those obtained at τ = 501.

[Please insert Table 3 here]

From Table 3, starting with k_M = 0.125, it becomes evident that for small values of λ_M, the ARL_0 is as low as 250, indicating a higher likelihood of false alarms for smaller values of k_M. As λ_M approaches 1, the ARL_0 steadily increases, reaching a peak of 334.8 at λ_M = 1. For ARL_1, the performance of MEC chart improves as λ_M increases. From Table 3, it is clear that λ_M = 1 produces the smallest ARL values for all shifts, particularly at lower k_M. However, this trend changes as k_M increases beyond 0.25. For k_M = 0.5 and larger, moderate values of λ_M (around 0.1) produce smaller ARL values.

[Please insert Tables 4-5 here]

It is also evident from Tables 4 and 5 that as both k_M and λ_M increase, the ARL_0 continues to improve, moving closer to the desired ARL_0 value of 370. This indicates that under steady-state conditions, the MEC chart exhibits a lower false alarm rate with larger design parameter values.

[Please insert Table 6 here]

From Tables 6 and 7, we observe that larger values of k_M (ranging from 1.5 to 2.5) tend to perform better with smaller values of λ_M. For instance, with k_M = 2 and λ_M = 0.025 (as shown in Table 6), only 26.89 samples are required on average to detect a shift of 0.5 standard deviations, whereas about 93 samples are needed for the same shift when λ_M = 0.75. Additionally, from Figure 1, it is noted that for larger values of k_M, different λ_M values exhibit nearly identical performance when detecting large shifts ($\delta \geq 3$).

[Please insert Table 7 and Figure 1 here]

Steady-State EQL Performance: As in the previous section, the overall performance of the MEC chart is analyzed under steady-state conditions using the EQL metric. The lower rows of Table 1 present the steady-state EQL values of MEC chart. The same trends observed under the zero-state condition apply here as well.

From Table 5, it is clear that nearly all EQL values below 20 occur when $k_M \geq 1$ and $\lambda_M \geq 0.5$. In general, larger values of the MEC design parameters are preferred when the goal is to optimize overall performance.

For a graphical representation, Figure 2 illustrates the effect of the parameters k_M and λ_M on the overall performance, as measured by EQL. Clearly, increasing the values of both parameters up to certain limits has a positive impact on the performance of MEC chart.

[Please insert Figure 2 here]

Steady-State AEL Performance: The steady-state AEL values of MEC chart are also provided in Table 2. As seen in the zero-state scenario, the largest AEL values occur when both k_M and λ_M are at their smallest values. By fixing λ_M, we also observe that AEL steadily decreases as k_M increases. For instance, the AEL value decreases from 62.9188 at k_M = 0.125 and λ_M = 0.01 to 18.3877 at k_M = 1.5 for the same value of λ_M.

Among all parameter combinations considered, the lowest AEL value of 8.3574 occurs when k_M = 2.5 and λ_M = 0.25. This value is slightly lower than the 8.389 observed under the zero-state condition.

4. Comparative Analysis

This section presents a detailed comparison between the MEC chart and two other memory-type control charts: the EWMA and CUSUM charts. Since the MEC chart combines features from both EWMA and CUSUM charts, we use the same parameter values for the comparisons. Various methods exist for comparing control charts, including direct comparisons of performance metrics such as the ARL or ensuring equivalent variance properties for the competing charts (for details, see [20]). However, to simplify the comparison process, we focus on the CED profiles and overall performance using the EQL and AEL metrics.

For a meaningful comparison, the performance of the MEC chart is evaluated based on both zero-state and steady-state conditions. In zero-state comparisons, the process starts under IC conditions, whereas in steady-state comparisons, we assume the process has been operating for some time, allowing for a deeper understanding of long-term performance. The metrics we examine include the EQL, AEL, and SEQL values for different process shifts. By comparing these metrics, we assess how well the charts detect both small and large shifts in the process mean, helping practitioners choose the best control chart depending on the situation.

4.1. Zero-State Comparisons

In this subsection, we compare the performance of the MEC, EWMA, and CUSUM charts based on zero-state conditions, where the process is assumed to start under IC conditions. We first focus on the EQL values, followed by the AEL metrics.

EQL Comparisons: Table 1 presents the zero-state EQL values for the MEC chart. The values less than 20 are highlighted in bold, and it is observed that the chart generally performs better for $k_M \geq 1$ and $\lambda_M \geq 0.375$. The minimum EQL value is attained when both k_M and λ_M are equal to 1, which suggests that the CUSUM chart with k_C = 1 performs slightly better than the MEC chart designs considered here. From Table 8, the lowest EQL value of EWMA chart is 19.8530 at λ_E = 0.5, which is slightly higher than the EQL of MEC chart at k_M = 1.5 and λ_M = 0.75. This indicates the superiority of MEC chart over the EWMA chart in this context. In summary, the order of performance based on EQL is: CUSUM, MEC, and EWMA, with CUSUM leading the way.

AEL Comparisons: Next, we turn to the AEL metric, which provides a different perspective on chart performance due to its unique weighing method. As with EQL, larger k_M values tend to result in smaller AEL values. However, unlike EQL, moderate values of λ_M seem to offer better performance. Table 2 presents the AEL values for the MEC chart. The minimum AEL value of 8.3895 occurs at λ_M = 0.25 and k_M = 2.5, with nearly all values close to 8 occurring at $k_M \geq$ 1.5. For the EWMA chart, the minimum AEL value is also at λ_E = 0.25 and is slightly lower than

the AEL of MEC chart (see Table 8). The minimum AEL value of CUSUM chart is 8.5750 at k_C = 0.75. Consequently, the order of performance based on AEL is: EWMA, MEC, and CUSUM. Thus, when comparing the charts based on EQL and AEL, the MEC chart consistently ranks second in performance, with CUSUM leading in EQL and EWMA performing best in AEL.

SEQL Comparisons: Finally, we examine the SEQL values to understand how the charts perform across varying shifts. Table 9 presents these values. For shifts up to δ = 1, the EWMA chart exhibits the best performance, with the MEC chart closely trailing. For shifts ranging from δ = 1.25 to δ = 4, the MEC chart generally performs better, although CUSUM and EWMA occasionally outperform it in specific scenarios. For δ > 4, the CUSUM chart dominates, although the MEC chart remains competitive.

[Please insert Tables 8-9 here]

In conclusion, the zero-state comparisons show that while no single chart consistently outperforms the others across all metrics, the MEC chart offers a good balance, particularly for moderate shifts. The choice of control chart should depend on the specific process shift magnitude and the focus of practitioner, whether it is minimizing EQL, AEL, or SEQL.

4.2. Steady-State Comparisons

In this subsection, we analyze and compare the performance of the MEC, EWMA, and CUSUM charts under steady-state conditions. In the steady-state, the process is allowed to reach its natural variability, simulating the long-term operation of a process.

EQL Comparisons: In Table 3, we presented the steady-state EQL values for the MEC chart. The performance trends are consistent with the zero-state results, where larger k_M values (greater than or equal to 1) result in better performance. The smallest EQL value is observed when k_M = 1 and λ_M = 0.75, with a value of 17.3321. This is comparable to the CUSUM chart, whose best steady-state EQL value is 17.2503 at k_C = 1. For the EWMA chart, the minimum steady-state EQL value is 18.1540 at λ_E = 0.25 (see Table 8). Therefore, as with the zero-state comparison, the CUSUM chart performs slightly better than the MEC and EWMA charts in terms of EQL, with the MEC chart ranking in second place. As shown in Figure 3, the MEC, EWMA, and CUSUM charts exhibit nearly identical performance across all shifts. This comparison is made under the condition that each chart is designed to achieve the minimum EQL value within its respective scheme.

AEL Comparisons: Moving to the AEL metric, the results in steady-state show a slightly different trend. Table 4 shows that the MEC chart achieves its minimum AEL value of 8.2345 at k_M = 2.0 and λ_M = 0.5. This value is very close to the minimum steady-state AEL value of the EWMA chart, which is 8.1748 at λ_E = 0.25 (see Table 8). On the other hand, the CUSUM chart has a minimum steady-state AEL value of 8.4532 at k_C = 1.25 (see Table 8). Thus, based on AEL, the order of performance in the steady-state scenario mirrors that of the zero-state comparison: EWMA, MEC, and CUSUM.

SEQL Comparisons: Next, we examine the SEQL values, which capture the dynamic behavior of the charts for various shifts over time. Table 9 presents the SEQL values for the different charts. As observed in the zero-state comparison, for small shifts ($\delta \leq 1.25$), the EWMA chart

maintains its superiority, with the MEC chart being a close contender. For moderate shifts, $1.5 \leq \delta \leq 4$, the MEC chart displays its strength by achieving lower SEQL values than both the EWMA and CUSUM charts. For $\delta > 4$, the CUSUM chart dominates the SEQL metric, though the MEC chart still performs reasonably well.

[Please insert Table 9 and Figure 3 here]

The steady-state comparisons reinforce the zero-state findings, with the MEC chart performing best for moderate shifts, while the CUSUM chart excels in detecting larger shifts, and the EWMA chart remains optimal for small shifts. The MEC chart provides a balanced alternative to the other two, making it a viable choice across different process shift scenarios.

- In terms of EQL, the CUSUM chart is marginally superior to the MEC and EWMA charts.
- Based on AEL, the EWMA chart outperforms both the MEC and CUSUM charts.
- For SEQL, the MEC chart shows excellent performance for moderate shifts, while the EWMA chart is best for small shifts and the CUSUM chart for large shifts.

Practitioners should select the control chart that aligns with their specific process monitoring needs, considering the expected magnitude of shifts and their focus on minimizing EQL, AEL, or SEQL.

4.3. CED Comparisons

This section investigates the CED for the MEC, EWMA, and CUSUM charts. The CED metric is used to evaluate the average number of observations needed to detect a shift, given that the shift occurs at a specific point in time. It provides a detailed understanding of the detection capabilities of chart at various shift magnitudes and moments after the chart becomes operational.

Performance for Small Shifts ($\delta \leq 0.5$): For small shifts, the CED values are significantly influenced by the memory parameter (λ_E) in the EWMA and MEC charts. Table 5 shows that the EWMA chart exhibits superior performance for small shifts when λ_E = 0.1, achieving the shortest CED values across all time intervals considered. The CUSUM chart, though highly efficient at detecting moderate to large shifts, struggles with small shifts, showing higher CED values in comparison to both EWMA and MEC charts. The MEC chart, with λ_M = 0.75 and k_M = 0.25, performs moderately for small shifts, offering a compromise between the faster detection ability of the EWMA chart and the slower response of the CUSUM chart. The CED values for the MEC chart are not as small as those of the EWMA chart but still considerably lower than those of the CUSUM chart, making it a reasonable alternative for detecting minor shifts.

Performance for Moderate Shifts ($0.5 < \delta \leq 2$): For moderate shifts, the strengths of MEC chart become more apparent. Table 6 presents the CED values for moderate shifts and shows that the MEC chart achieves better performance than the EWMA chart for δ between 0.75 and 1.75. The CUSUM chart, while still competitive, begins to lag behind the MEC chart as the shift size decreases. At λ_M = 0.75 and k_M = 1, the MEC chart demonstrates consistently lower CED values across different time intervals. This makes it particularly suitable for processes

where moderate shifts are expected to occur. The CUSUM chart, although capable, tends to detect these shifts slightly later than the MEC chart, while the EWMA chart shows the largest delay for these shifts.

Performance for Large Shifts (δ > 2): For larger shifts, the CUSUM chart clearly outperforms both the MEC and EWMA charts in terms of CED. As shown in Table 7, the CUSUM chart with k_C = 1 consistently detects large shifts (δ > 2) faster than the other charts. This performance is expected due to the structure of CUSUM, which is highly sensitive to large, sustained deviations from the process mean. The MEC chart also performs well for large shifts, particularly when k_M is set to ≥ 1.5. However, it does not match the speed of detection offered by the CUSUM chart. The EWMA chart, with its memory structure optimized for smaller shifts, shows the longest detection delays for large shifts, making it less effective in this context.

CED Profiles Across Time: Figures 4 to 7 illustrate the CED profiles for the MEC, EWMA, and CUSUM charts across different shift magnitudes and time intervals. The CED profiles provide a visual representation of how quickly each chart detects shifts over time. For small shifts (Figures 4 and 5), the EWMA chart dominates, showing the fastest detection times, while the performance of MEC chart remains satisfactory. For moderate shifts (Figures 5 and 6), the MEC chart takes the lead, providing faster detection than the EWMA chart, and approaches the performance of CUSUM chart. Finally, for large shifts (Figure 7), the CUSUM chart remains superior in terms of detection speed, with the MEC chart performing similarly but slightly slower.

[Please insert Figures 4-7 here]

To summarize:

- For small shifts (δ ≤ 0.5), the EWMA chart offers the best performance, with the MEC chart as a viable alternative, outperforming the CUSUM chart.
- For moderate shifts (0.5 < δ ≤ 2), the MEC chart demonstrates superior performance compared to both the EWMA and CUSUM charts, making it ideal for processes prone to moderate deviations.
- For large shifts (δ > 2), the CUSUM chart remains the best option, although the MEC chart performs reasonably well.
- The CED profiles over time reflect these trends, with the EWMA chart dominating for small shifts, the MEC chart excelling in moderate shift detection, and the CUSUM chart performing best for large shifts.

In practice, the choice of the control chart should depend on the expected magnitude of process shifts and the need to minimize detection delay at different points in time.

4.4. Design of MEC Chart

In some cases, the practitioner may want to optimize the control chart to detect a specific shift with the minimum ARL. To this end, we provide the steady-state optimal values of the pair (k_M, λ_M) for each anticipated shift size (δ_{Target}) and their corresponding control limit h_M coefficient in Table 10. For a specified δ_{Target}, the unique combination (k_M, λ_M, h_M) gives the

minimum ARL_1 (ARL*). It is observed that for all the different shift magnitudes presented here, except for δ_{Target} = 0.25, the MEC chart is optimized at λ_M = 0.75 with different values of k_M depending on the size of the shift. For $\delta_{\text{Target}} \geq 3.5$, the optimal setting is k_M = 2.5 and λ_M = 0.75.

Using the information from Table 10 and the 3D plot in Figure 8, we summarize the relationship among the parameters, magnitude of shifts, and the optimal ARL as follows:

- For almost all $\delta_{\text{Target}} \geq 0.5$, λ_M remains constant at 0.75, suggesting this value provides a reliable optimization parameter across a wide range of shifts. This simplifies the control chart configuration since λ_M does not need to be adjusted frequently. The exception is δ_{Target} = 0.25, where λ_M = 0.5, indicating that smaller shifts may require a different value for λ_M to optimize the ARL*.
- k_M increases with increasing δ_{Target}, but shows a nonlinear relationship:
    - At low target shifts (e.g., δ_{Target} = 0.25 to δ_{Target} = 0.75), k_M changes from 0.125 to 0.5.
    - For medium target shifts (e.g., δ_{Target} = 1 to δ_{Target} = 2.5), k_M ranges from 0.5 to 1.5.
    - For higher target shifts ($\delta_{\text{Target}} \geq 2.75$), k_M reaches 2 or higher.
- This gradual increase suggests that as the magnitude of the shift increases, a larger value of k_M is necessary to detect it efficiently with minimal ARL.
- The control limit h_M decreases as δ_{Target} increases:
    - At δ_{Target} = 0.25, h_M = 24.72, indicating that a larger control limit is required for smaller shifts.
    - At δ_{Target} = 4, h_M = 0.505, indicating a much smaller control limit for larger shifts.
- This trend shows that larger shifts require tighter control limits to detect OOC situations quickly.
- ARL* decreases dramatically with increasing δ_{Target}:
    - For δ_{Target} = 0.25, ARL* is 62.30, meaning it takes more time to detect small shifts.
    - For larger shifts, such as δ_{Target} = 4, ARL* drops to 1.20, meaning the process can detect large shifts quickly.

For δ_{Target} = 0.25, both λ_M and k_M take lower values (0.5 and 0.125, respectively), indicating a unique optimization for small shifts. This is in contrast to larger shifts, where k_M and λ_M stabilize at higher values.

At higher shift magnitudes ($\delta_{\text{Target}} \geq 3.5$), the combination of k_M = 2.5 and λ_M = 0.75 remains constant, while h_M reduces significantly. This suggests the process stabilizes at a particular combination for large shifts, optimizing detection without further parameter changes.

The relationship between k_M and h_M is nonlinear, as seen in the 3D plot in Figure 8. It can be observed that while k_M increases steadily with δ_{Target}, the corresponding decrease in h_M is

much steeper, especially at higher shift values. Figure 8 can be used as a guide to determine the value of the control limit h_M for any pair given optimal pairs k_M and λ_M.

[Please insert Table 10 and Figure 8 here]

The above analysis indicates that practitioners must carefully balance these parameters based on the anticipated shift magnitude to ensure efficient detection. Understanding the interaction between k_M, λ_M, and h_M is crucial for minimizing ARL. For smaller shifts, lower values of k_M and λ_M require larger control limits h_M, while for more significant shifts, higher k_M with a stable λ_M and smaller h_M achieve the desired ARL.

5. Real-Time Monitoring of Water pH Levels in Aquatic Ecotoxicology

This section illustrates the practical application of the MEC chart, alongside the CUSUM and EWMA charts, for monitoring water pH levels in an aquatic ecotoxicology laboratory. The goal is to assess the performance of these control charts in detecting changes in water pH, a critical factor in ensuring optimal living conditions for aquatic organisms in the lab. The laboratory under study maintains a controlled environment for mysids, small crustaceans used in water toxicity assessments. Consistent monitoring of water pH is crucial to maintaining the health of these organisms and ensuring the accuracy of ecotoxicological experiments. In this application, we aim to detect any deviations from the target pH level of 8.2, which could indicate a potential issue in the water quality management system.

To achieve this, the MEC chart is applied alongside traditional control charts like the CUSUM, EWMA, and Shewhart charts. A fair comparison is ensured by setting the parameters of all charts such that their ARL_0 is 370. The control charts are then used to analyze the pH data, which includes both an IC period and an OOC period caused by scheduled maintenance in the laboratory, leading to a known disruption in water circulation. Further details about the data, assumptions, and control chart setup are provided in the following subsections.

5.1. pH Data Description

The ecotoxicology laboratory is designed to maintain a controlled setting for investigating the harmful effects of chemicals on aquatic life, particularly focusing on mysids. These small crustaceans, measuring approximately 1 cm in length, inhabit both fresh and saltwater environments and are frequently used as indicators in water toxicity assessments. The laboratory houses mysids in tanks filled with either fresh or saltwater, and the water undergoes continuous circulation and filtration for 22 hours daily to ensure cleanliness and prevent algae growth. The laboratory supervisor provides food for the mysids twice daily.

The facility maintains specific water conditions critical for mysid health, including a temperature of 23.8°C (75°F), illumination at 941.775 LUX, a pH level of 8.2, an ammonia (NH3) concentration of 0.1 mg/L, and salinity ranging from 20 to 22 parts per thousand (ppt). These controlled conditions are essential for safeguarding the health of the mysids and ensuring reliable experimental outcomes. For more detailed information about the lab setting, refer to [29]. The data collected for this study consists of daily pH measurements

taken over a period of 60 days, from April 1 to May 30, 2016, which are reproduced in Table 11.

[Please insert Table 11 here]

5.2. Evaluating Key Statistical Assumptions

Before applying the control charts, it is crucial to validate the underlying assumptions concerning the pH data, particularly focusing on its normality and independence. The normality assumption is foundational in SPC, as many control chart methodologies rely on the data being normally distributed. To assess this, we employed the Anderson-Darling normality test (see [30]). The test yielded a p-value of 0.1191, indicating that there is no significant departure from normality. In addition to normality, we also evaluated the independence of observations, which is essential for ensuring that the control chart signals are reliable and not influenced by autocorrelation. The Durbin-Watson test (see [31]) was utilized for this purpose, yielding a p-value of 0.902. This high p-value indicates a lack of significant autocorrelation among the pH measurements, confirming that the observations are independent. Thus, the independence assumption holds, allowing us to proceed with the application of the control charts without the risk of misleading results due to correlated data.

5.3. Monitoring pH Level Using Control Charts

We apply the three control charts discussed in this research, along with the Shewhart chart, to the pH data provided in Table 11. According to [29], the IC mean and standard deviation of the water pH are 8.2 and 0.1, respectively. The control charts are established using these known values, with the following charting constants for each chart:

- **Shewhart Chart:** Control limit coefficient L_S = 3
- **EWMA Chart:** Sensitivity parameter λ_E = 0.02, asymptotic control limit coefficient L_E = 2.1349
- **CUSUM Chart:** Sensitivity parameter k_C = 0.125 and control limit coefficient h_C = 12.0829
- **MEC Chart:** Sensitivity parameters k_M = 0.125, λ_M = 0.5, and control limit coefficient h_M = 24.72

These charting parameters and constants are selected for each chart so that the ARL_0 = 370, allowing for a fair comparison among the four charts.

The IC region (light green in Figure 10) encompasses the initial 50 days, while the OOC region (light red) spans from day 51 to 60. During the final 10-day period (days 51-60), the laboratory underwent annual maintenance, necessitating the temporary disconnection of an air supply pipe. This disruption affected water circulation, leading to comparatively elevated pH levels.

Upon analyzing Figure 10, it becomes evident that the performance of the control charts varies considerably. The Shewhart chart, despite its simplicity and ease of use, demonstrates limited sensitivity to small shifts in process parameters. This characteristic may render it less effective in detecting early signs of instability, especially in sensitive laboratory environments where maintaining a specific pH level is crucial for the health of mysids. The EWMA chart,

known for its sensitivity to shifts, signaled four false alarms starting from day 35, indicating an increased responsiveness to minor fluctuations in pH levels. However, this heightened sensitivity came at the expense of generating misleading signals, as evidenced by the fact that the first valid OOC signal from the EWMA chart occurred on day 53. In contrast, both the CUSUM and MEC charts issued their first valid OOC signal on day 54, demonstrating their capacity to accurately detect the shift without any false alarms. This is particularly significant, as it highlights the advantages of using these charts in scenarios where maintaining a high level of accuracy in monitoring is critical.

[Please insert Figure 10 here]

Overall, the performance comparison reveals that while the EWMA chart excels in early detection of OOC conditions, it is less effective in maintaining IC performance due to the numerous false alarms generated. The MEC and CUSUM charts exhibited a more balanced performance, effectively detecting shifts in pH levels with no false alarms, thereby providing reliable monitoring. This capability is particularly crucial in an ecotoxicology laboratory, where timely detection of water quality deviations is essential for the health of sensitive aquatic organisms, such as mysids. The findings highlight the necessity of selecting appropriate control charts tailored to the specific monitoring context, balancing sensitivity with the risk of false alarms. In this case, the MEC chart not only offers enhanced performance in detecting shifts but also minimizes the likelihood of erroneous signals, making it a valuable tool for real-time water pH regulation in aquatic environments.

6. Summary, Conclusions, and Recommendations

SPC remains a cornerstone technique for achieving and maintaining process stability, ensuring the consistent production of high-quality goods and services. Control charts, in particular, are powerful tools for detecting assignable causes of variation in processes and are widely recognized for their simplicity in implementation. This study focuses on the development and evaluation of the MEC control chart, a memory-type control chart that combines the strengths of EWMA and CUSUM methods to detect small to moderate shifts in process parameters more effectively than traditional Shewhart charts. Unlike previous studies that mainly assessed the performance of control charts using zero-state run length distribution, this research has offered a more comprehensive evaluation by considering both zero-state and steady-state properties. Through extensive Monte Carlo simulations, we evaluated the performance of the MEC chart using key metrics such as ARL, EQL and AEL. The results consistently show that the MEC chart demonstrates excellent performance across a wide range of shift magnitudes, particularly for moderate shifts where it excels in timely detection while maintaining a low false alarm rate. Additionally, the study has provided practitioners with a table of optimal parameters for the MEC chart, tailored for various shift magnitudes. This practical guide simplifies the process of configuring the control chart for real-world applications, enabling more efficient monitoring of process parameters and improving overall quality control.

The practical utility of the MEC chart was further validated through an application in an aquatic ecotoxicology laboratory, where it was used to monitor water pH levels. The laboratory, which maintains controlled environmental conditions for the study of aquatic organisms, especially mysids, relies on precise water quality to ensure the validity of experimental outcomes. Our study demonstrated that the MEC chart not only detected deviations in pH levels accurately but also outperformed the EWMA and Shewhart charts by minimizing false alarms and providing timely signals for process deviations during the maintenance period of laboratory. The sensitivity of MEC chart to moderate shifts, combined with its ability to maintain stability during IC conditions, makes it an invaluable tool for real-time monitoring in critical environments where precision is paramount.

In terms of practical applications, integration of MEC chart with automated monitoring systems would enhance real-time decision-making, reducing the need for manual interventions and improving process efficiency. Furthermore, beyond laboratory settings, the MEC chart could be adapted for broader industrial use, particularly in sectors that demand high precision and quality control, such as chemical manufacturing, electronics, and environmental monitoring. This adaptation would help improve the detection of small shifts in processes that rely heavily on stable parameter regulation.

Table 1. Zero- and steady-state EQL values of MEC control chart.

Zero-state									
	k_M								
λ_M	0.125	0.25	0.5	0.75	1	1.5	2	2.5	3
0.01	267.9132	239.0157	33.9723	146.9892	112.1147	61.0010	*	*	*
0.025	209.4027	187.3235	151.2752	122.9308	99.6575	63.9854	41.0289	*	*
0.05	165.0904	146.0910	117.8222	96.9490	80.4818	56.3164	38.9433	*	*
0.1	126.4136	109.5690	87.4637	72.5641	61.1478	45.2213	33.5855	28.4067	*
0.175	100.8031	85.3668	66.5829	55.6337	47.0487	35.3674	29.0487	24.0390	*
0.25	86.8803	72.2484	55.9796	46.1856	39.9217	31.1321	25.0454	21.6455	*
0.375	73.1363	59.3328	45.1806	36.8806	31.9678	26.4421	21.1334	**19.9595**	*
0.5	64.4493	50.5329	37.9541	31.6312	28.8741	22.2003	**19.7598**	**19.6970**	*
0.75	53.1480	40.6349	30.6360	25.6712	21.8358	**19.5835**	20.2607	21.3659	*
1	46.3701	33.9723	24.9039	20.5922	**19.5808**	21.1758	24.1427	26.2089	26.7919
Steady-state									
0.01	226.6431	207.9915	173.4082	139.9925	109.3752	60.3999	*	*	*
0.025	174.1998	161.1018	137.7157	116.1950	96.3152	63.3561	40.8457	*	*
0.05	135.6649	125.1107	107.2736	91.6542	77.8678	55.3115	38.8427	*	*
0.1	103.1400	94.1180	79.8700	68.5422	59.1317	44.3433	33.4553	27.4662	*
0.175	81.9707	73.5404	61.3297	52.7330	45.7726	35.3198	28.2183	23.9503	*
0.25	70.5654	62.3648	51.4180	43.9943	38.6101	30.8111	24.7808	21.9134	*
0.375	59.3849	51.3294	51.4180	35.3760	31.2606	25.9396	21.2907	20.2013	*
0.5	52.4251	44.2388	35.2568	30.3935	27.7506	22.1311	**19.8507**	**19.7933**	*
0.75	43.4774	35.3843	28.3184	24.3274	21.3034	**19.4754**	20.2276	21.3276	*
1	37.5910	29.8393	22.8436	**19.8058**	**19.1996**	21.0432	24.0654	26.1351	26.7304

Table 2. Zero- and steady-state AEL values of MEC control chart.

Zero-state									
	k_M								
λ_M	0.125	0.25	0.5	0.75	1	1.5	2	2.5	3
0.01	74.4776	66.5821	11.1971	41.5935	32.2195	18.5923	*	*	*
0.025	58.6200	52.5198	42.6172	34.8969	28.5967	19.0688	13.0602	*	*
0.05	46.6453	41.3156	33.4684	27.7676	23.3146	16.8296	12.3487	*	*
0.1	36.2516	31.4355	25.1927	21.0907	18.0333	13.8163	10.8731	9.5031	*
0.175	29.4025	24.8954	19.5840	16.5400	14.3074	11.3880	9.7938	8.6683	*
0.25	25.6954	21.3522	16.6977	14.0673	12.4561	10.3441	9.0540	**8.3895**	*
0.375	22.0348	17.8768	13.8325	11.7329	10.5538	9.4576	8.5748	8.5828	*
0.5	19.7191	15.5816	12.0256	10.4395	9.8548	8.8625	8.7988	9.2916	*
0.75	16.7048	12.9346	10.1645	9.2225	8.7823	9.2958	10.6953	11.8799	*
1	14.8033	11.1971	9.0145	8.5724	9.1750	11.9529	14.9576	16.5904	16.9782
	Steady-state								
0.01	62.9188	57.8840	48.5690	39.5837	31.3821	18.3877	*	*	*
0.025	48.6697	45.0998	38.7545	32.9421	27.5988	18.8198	12.9357	*	*
0.05	38.2144	35.3063	30.4304	26.1970	22.4927	16.5051	12.2087	*	*
0.1	29.4287	26.9038	22.9707	19.9033	17.3947	13.5332	10.7632	9.2436	*
0.175	23.7538	21.3576	17.9721	15.6569	13.8609	11.2622	9.5550	8.5594	*
0.25	20.7180	18.3585	15.3159	13.3692	12.0284	10.1964	8.9177	8.3574	*
0.375	17.7557	15.4127	15.3159	11.2035	10.2781	9.2773	8.5325	8.5658	*
0.5	15.9199	13.5476	11.1357	10.0029	9.5167	8.7734	8.7578	9.2554	*
0.75	13.5621	11.2384	9.4220	8.7896	8.5712	9.2170	10.6494	11.8312	*
1	11.9974	9.8082	8.3396	8.2579	8.9965	11.8740	14.8999	16.5407	16.9332

Table 3. Steady-state ARL profiles of MEC control chart with zero-state $ARL_0 \cong 370$, $k_M = 0.125$ and τ = 201.

	λ_M									
δ	0.01	0.025	0.05	0.1	0.175	0.25	0.375	0.5	0.75	1
0	250.75	261.49	268.45	277.10	287.36	295.32	305.23	313.14	325.92	334.80
0.25	87.10	76.65	70.09	65.80	63.73	62.93	62.32	62.30	62.62	62.10
0.5	57.37	47.30	40.23	34.84	31.88	30.44	29.06	28.18	26.96	26.10
0.75	45.50	36.66	30.23	25.08	22.05	20.58	19.17	18.29	17.08	16.14
1	38.78	30.82	25.02	20.27	17.38	15.89	14.52	13.68	12.54	11.70
1.25	34.29	27.06	21.75	17.36	14.59	13.16	11.83	11.02	9.97	9.19
1.5	31.05	24.36	19.44	15.34	12.74	11.38	10.09	9.32	8.31	7.59
1.75	28.61	22.33	17.73	13.87	11.40	10.10	8.86	8.11	7.16	6.49
2	26.61	20.71	16.37	12.73	10.39	9.15	7.96	7.23	6.31	5.67
2.25	25.00	19.40	15.29	11.82	9.59	8.40	7.25	6.54	5.65	5.06
2.5	23.63	18.30	14.38	11.08	8.94	7.79	6.68	6.00	5.14	4.57
2.75	22.47	17.37	13.61	10.45	8.39	7.30	6.22	5.56	4.73	4.17
3	21.46	16.56	12.95	9.91	7.93	6.88	5.84	5.20	4.39	3.85
3.25	20.57	15.86	12.38	9.44	7.54	6.51	5.51	4.89	4.10	3.58
3.5	19.79	15.22	11.87	9.04	7.19	6.20	5.22	4.62	3.85	3.35
3.75	19.08	14.67	11.43	8.68	6.89	5.93	4.98	4.39	3.64	3.14
4	18.46	14.17	11.02	8.35	6.62	5.68	4.76	4.19	3.45	2.97
4.25	17.88	13.72	10.66	8.07	6.38	5.47	4.57	4.01	3.30	2.82
4.5	17.36	13.31	10.33	7.80	6.16	5.27	4.39	3.85	3.15	2.69
4.75	16.87	12.91	10.01	7.57	5.96	5.09	4.24	3.70	3.02	2.57
5	16.42	12.57	9.74	7.35	5.78	4.93	4.10	3.57	2.90	2.47
5.25	16.01	12.24	9.48	7.14	5.61	4.79	3.97	3.45	2.79	2.38
5.5	15.63	11.94	9.24	6.95	5.46	4.65	3.86	3.35	2.69	2.29
5.75	15.26	11.67	9.01	6.78	5.32	4.52	3.74	3.26	2.62	2.19
6	14.94	11.41	8.82	6.62	5.18	4.40	3.63	3.17	2.55	2.11

Table 4. Steady-state ARL profiles of MEC control chart with zero-state $ARL_0 \cong 370$, $k_M = 0.25$ and τ = 201.

	λ_M									
δ	0.01	0.025	0.05	0.10	0.175	0.25	0.375	0.50	0.75	1
0	268.61	283.01	293.71	306.03	318.57	326.74	335.12	340	348.69	354.59
0.25	83.88	74.2	68.1	64.41	63.14	63.16	63.97	65.54	70.52	77.53
0.5	54.05	44.7	37.87	32.52	29.47	27.95	26.64	25.89	25.34	25.66
0.75	42.51	34.38	28.21	23.1	19.97	18.39	16.9	15.98	14.87	14.27
1	36.09	28.81	23.26	18.58	15.61	14.05	12.58	11.69	10.56	9.85
1.25	31.81	25.19	20.16	15.88	13.08	11.6	10.18	9.31	8.23	7.53
1.5	28.74	22.66	18.01	14.02	11.41	10.01	8.65	7.83	6.78	6.11
1.75	26.41	20.74	16.4	12.67	10.21	8.89	7.59	6.8	5.81	5.16
2	24.55	19.23	15.14	11.63	9.3	8.04	6.81	6.05	5.1	4.49
2.25	23.04	17.99	14.12	10.79	8.59	7.39	6.21	5.48	4.56	3.97
2.5	21.76	16.96	13.28	10.11	8	6.86	5.73	5.03	4.15	3.58
2.75	20.68	16.09	12.56	9.54	7.52	6.42	5.34	4.66	3.81	3.26
3	19.72	15.34	11.95	9.04	7.11	6.05	5.02	4.36	3.54	3.01
3.25	18.91	14.67	11.42	8.61	6.76	5.74	4.73	4.1	3.31	2.8
3.5	18.16	14.08	10.95	8.25	6.45	5.47	4.5	3.89	3.11	2.62
3.75	17.52	13.56	10.54	7.91	6.18	5.23	4.29	3.7	2.94	2.46
4	16.93	13.1	10.16	7.62	5.94	5.02	4.11	3.53	2.8	2.32
4.25	16.39	12.68	9.82	7.36	5.72	4.83	3.94	3.39	2.68	2.2
4.5	15.91	12.29	9.51	7.12	5.52	4.65	3.79	3.26	2.58	2.09
4.75	15.46	11.94	9.24	6.9	5.34	4.5	3.66	3.13	2.48	2
5	15.04	11.61	8.97	6.7	5.18	4.36	3.54	3.01	2.38	1.93
5.25	14.66	11.31	8.73	6.52	5.04	4.24	3.44	2.91	2.28	1.87
5.5	14.31	11.03	8.51	6.34	4.9	4.12	3.35	2.82	2.19	1.82
5.75	13.98	10.77	8.31	6.18	4.78	4.01	3.26	2.75	2.1	1.79
6	13.67	10.52	8.11	6.04	4.65	3.9	3.17	2.69	2.03	1.75

Table 5. Steady-state ARL profiles of MEC control chart with zero-state $ARL_0 \cong 370$, $k_M = 0.5$ and τ = 201.

	λ_M									
δ	0.01	0.025	0.05	0.10	0.175	0.25	0.375	0.50	0.75	1
0	297.21	310.34	324.02	334.74	343.85	348.69	348.69	356.22	361.08	363.04
0.25	78.55	70.63	66.2	64.8	66.51	69.44	69.44	82.45	98.09	118.38
0.5	48.11	40.32	34.25	29.44	26.9	25.98	25.98	26.21	28.78	33.61
0.75	37.1	30.44	24.98	20.28	17.35	15.96	15.96	14.28	14.22	15.15
1	31.13	25.27	20.4	16.1	13.31	11.87	11.87	9.87	9.22	9.18
1.25	27.25	21.98	17.59	13.64	11.05	9.68	9.68	7.64	6.84	6.52
1.5	24.49	19.69	15.64	12.01	9.59	8.29	8.29	6.32	5.48	5.06
1.75	22.4	17.95	14.19	10.82	8.56	7.34	7.34	5.44	4.6	4.15
2	20.74	16.6	13.08	9.91	7.78	6.63	6.63	4.82	4	3.53
2.25	19.41	15.49	12.18	9.19	7.17	6.08	6.08	4.35	3.56	3.09
2.5	18.29	14.58	11.44	8.6	6.68	5.64	5.64	3.99	3.22	2.75
2.75	17.34	13.81	10.81	8.1	6.27	5.28	5.28	3.7	2.95	2.49
3	16.52	13.14	10.27	7.68	5.93	4.97	4.97	3.45	2.73	2.29
3.25	15.81	12.57	9.8	7.32	5.63	4.72	4.72	3.25	2.55	2.12
3.5	15.17	12.05	9.4	6.99	5.38	4.49	4.49	3.07	2.4	1.99
3.75	14.61	11.6	9.03	6.71	5.15	4.3	4.3	2.93	2.26	1.89
4	14.1	11.19	8.7	6.46	4.94	4.12	4.12	2.81	2.14	1.79
4.25	13.63	10.82	8.4	6.23	4.76	3.96	3.96	2.7	2.05	1.71
4.5	13.22	10.48	8.14	6.03	4.6	3.82	3.82	2.61	1.98	1.62
4.75	12.84	10.17	7.9	5.84	4.46	3.7	3.7	2.51	1.92	1.54
5	12.49	9.89	7.67	5.67	4.32	3.6	3.6	2.42	1.88	1.46
5.25	12.16	9.63	7.46	5.51	4.2	3.5	3.5	2.32	1.85	1.37
5.5	11.86	9.39	7.27	5.37	4.07	3.4	3.4	2.22	1.81	1.29
5.75	11.58	9.16	7.1	5.24	3.96	3.3	3.3	2.14	1.78	1.22
6	11.32	8.96	6.93	5.11	3.85	3.2	3.2	2.06	1.74	1.16

Table 6. Steady-state ARL profiles of MEC control chart with zero-state $ARL_0 \cong 370$, $k_M = 2$ and τ = 201.

	λ_M								
δ	0.025	0.05	0.10	0.175	0.25	0.375	0.50	0.75	1
0	346.84	353.78	360.44	363.11	364.58	367.04	367.99	367.72	368.36
0.25	64.28	69.81	83.92	103.9	122.07	149.57	174.99	225.33	272.77
0.5	26.89	25.87	26.98	30.92	36.12	46.4	58.81	93.01	142.97
0.75	16.82	15.4	14.52	14.87	16.1	19.42	24.13	40.16	70.82
1	12.32	11.09	9.93	9.42	9.54	10.52	12.3	19.52	36.25
1.25	9.77	8.77	7.66	6.92	6.68	6.84	7.5	10.8	19.65
1.5	8.13	7.32	6.3	5.55	5.18	5.03	5.22	6.75	11.42
1.75	6.99	6.31	5.4	4.68	4.29	4	3.97	4.64	7.13
2	6.15	5.58	4.76	4.07	3.69	3.34	3.21	3.47	4.8
2.25	5.5	5.02	4.28	3.63	3.26	2.9	2.72	2.75	3.45
2.5	4.98	4.58	3.9	3.3	2.93	2.57	2.37	2.29	2.65
2.75	4.55	4.22	3.59	3.03	2.68	2.32	2.11	1.97	2.12
3	4.21	3.93	3.34	2.81	2.47	2.12	1.91	1.73	1.79
3.25	3.91	3.67	3.13	2.63	2.3	1.96	1.75	1.55	1.55
3.5	3.66	3.46	2.95	2.47	2.16	1.83	1.62	1.41	1.38
3.75	3.44	3.27	2.8	2.33	2.03	1.71	1.5	1.3	1.26
4	3.25	3.11	2.66	2.21	1.92	1.61	1.41	1.21	1.17
4.25	3.08	2.96	2.54	2.1	1.83	1.52	1.32	1.15	1.11
4.5	2.93	2.83	2.42	2.01	1.74	1.43	1.24	1.1	1.07
4.75	2.8	2.7	2.32	1.92	1.67	1.35	1.18	1.06	1.04
5	2.68	2.6	2.23	1.85	1.59	1.28	1.13	1.04	1.02
5.25	2.57	2.49	2.15	1.78	1.52	1.22	1.09	1.02	1.01
5.5	2.48	2.4	2.07	1.72	1.45	1.16	1.06	1.01	1
5.75	2.39	2.32	2	1.66	1.39	1.12	1.03	1	1
6	2.31	2.24	1.94	1.6	1.32	1.08	1.02	1	1

Table 7**.** Steady-state ARL profiles of MEC control chart with zero-state $ARL_0 \cong 370$, $k_M = 2.5$ & 3 and τ = 201.

	λ_M							
δ	0.1	0.175	0.25	0.375	0.5	0.75	1	1 (k=3)
0	360.9	363.72	365.42	366.41	369.26	368.23	368.92	369.49
0.25	86.74	109.96	130.92	162.55	191.36	239.64	279.09	280.09
0.5	27.31	32.69	39.45	52.55	68.34	106.35	153.19	154.86
0.75	14.31	15.28	17.22	21.89	28.42	48.13	79.48	80.98
1	9.52	9.37	9.84	11.57	14.31	24	42.49	43.69
1.25	7.12	6.68	6.69	7.27	8.46	13.29	23.95	24.88
1.5	5.71	5.2	5.04	5.17	5.67	8.13	14.24	14.94
1.75	4.79	4.28	4.06	3.99	4.18	5.43	8.92	9.44
2	4.14	3.65	3.41	3.26	3.29	3.92	5.92	6.28
2.25	3.66	3.2	2.96	2.77	2.72	3	4.15	4.4
2.5	3.29	2.86	2.63	2.42	2.33	2.42	3.06	3.24
2.75	3	2.6	2.37	2.16	2.05	2.03	2.37	2.48
3	2.76	2.38	2.17	1.96	1.84	1.76	1.92	2
3.25	2.56	2.21	2.01	1.8	1.67	1.55	1.62	1.67
3.5	2.39	2.06	1.87	1.67	1.54	1.4	1.41	1.44
3.75	2.26	1.94	1.76	1.56	1.42	1.29	1.28	1.29
4	2.13	1.84	1.66	1.46	1.33	1.2	1.18	1.19
4.25	2.03	1.75	1.58	1.38	1.25	1.13	1.11	1.11
4.5	1.94	1.67	1.5	1.3	1.18	1.09	1.07	1.07
4.75	1.85	1.6	1.42	1.23	1.13	1.05	1.04	1.04
5	1.78	1.53	1.36	1.18	1.09	1.03	1.02	1.02
5.25	1.72	1.47	1.3	1.13	1.06	1.02	1.01	1.01
5.5	1.66	1.41	1.24	1.09	1.04	1.01	1	1
5.75	1.6	1.35	1.19	1.06	1.02	1	1	1
6	1.55	1.3	1.15	1.04	1.01	1	1	1

Table 8. EWMA and CUSUM overall performance under both zero- and steady-state conditions using EQL and AEL metrics

EWMA					CUSUM				
	Zero-state		Steady-state			Zero-state		Steady-state	
λ_E	EQL	AEL	EQL	AEL	k_C	EQL	AEL	EQL	AEL
0.01	47.9397	15.1407	47.4117	37.591	0.125	46.3217	14.7884	37.591	11.9974
0.025	38.7397	12.4472	38.7579	29.8393	0.25	33.9741	11.1987	29.8393	9.8082
0.05	32.9309	10.7152	32.5849	22.8436	0.5	24.9044	9.0156	22.8436	8.3396
0.1	28.2043	9.4534	27.2624	19.8058	0.75	20.5995	**8.575**	19.8058	**8.2579**
0.175	23.6592	8.6006	23.66	**19.1996**	**1**	**19.5742**	9.1734	**19.1996**	8.9965
0.25	21.3753	**8.376**	21.6975	21.0432	1.5	21.1803	11.9557	21.0432	11.874
0.375	19.9223	8.6902	20.1824	24.0654	2	24.1269	14.9412	24.0654	14.8999
0.5	**19.853**	9.491	**19.9149**	26.1351	2.5	26.2144	16.5988	26.1351	16.5407
0.75	21.7631	12.2247	21.7323	26.7304	3	26.798	16.9801	26.7304	16.9332
1	26.8056	16.9915	26.7367	37.591					

Table 9. Zero- and steady state minimum Sequential EQL values of MEC, CUSUM, and EWMA control charts.

	Zero-state			Steady-state		
δ	MEC	CUSUM	EWMA	MEC	EWMA	CUSUM
0.5	5.5647	6.2194	5.5400	5.266	5.353	5.266
0.75	6.7069	7.2131	6.6027	6.360	6.406	6.426
1	7.6552	8.2460	7.6290	7.263	7.421	7.263
1.25	8.3791	9.2967	8.3851	8.149	8.171	8.149
1.5	9.1485	9.7872	9.1474	8.866	8.927	9.071
1.75	9.9295	10.3471	9.9481	9.590	9.721	9.622
2	10.5276	10.9627	10.5447	10.164	10.329	10.164
2.25	11.1700	11.6237	11.1679	10.753	10.946	10.753
2.5	11.8490	12.3216	11.8292	11.383	11.604	11.383
2.75	12.4167	12.7993	12.4523	12.045	12.222	12.045
3	13.0280	13.2779	13.0377	12.660	12.791	12.66
3.25	13.6814	13.8046	13.6659	13.148	13.399	13.148
3.5	14.2968	14.3701	14.3345	13.672	14.044	13.672
3.75	14.7891	14.9605	14.9078	14.224	14.676	14.224
4	15.2977	15.4036	15.3932	14.795	15.187	14.795
4.25	15.7686	15.7951	15.8862	15.366	15.726	15.366
4.5	16.2193	16.2103	16.3915	15.783	16.285	15.783
4.75	16.6664	16.6579	16.9116	16.231	16.865	16.231
5	17.1523	17.1443	17.4465	16.721	17.472	16.721
5.25	17.6799	17.6724	17.9929	17.257	18.026	17.257
5.5	18.2552	18.2480	18.5533	17.846	18.598	17.846
5.75	18.8856	18.8787	19.1702	18.495	19.227	18.495
6	19.5808	19.5742	19.8530	19.200	19.915	19.2

Table 10. The MEC chart optimal values of k_M and λ_M and corresponding limit h_M for target shifts. τ = 201.

δ	k_M	λ_M	h_M	ARL*
0.25	0.125	0.5	24.72	62.30
0.5	0.25	0.75	11.47	25.34
0.75	0.5	0.75	6.815	14.22
1	0.5	0.75	6.815	9.22
1.25	0.75	0.75	4.650	6.50
1.5	1	0.75	3.360	4.92
1.75	1	0.75	3.360	3.92
2	1.5	0.75	1.870	3.18
2.25	1.5	0.75	1.870	2.65
2.5	1.5	0.75	1.870	2.28
2.75	2	0.75	1.068	1.97
3	2	0.75	1.068	1.73
3.25	2	0.75	1.068	1.55
3.5	2.5	0.75	0.504	1.4
3.75	2.5	0.75	0.504	1.29
4	2.5	0.75	0.505	1.2

Table 11. pH levels Data

Days	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
pH value	8.2	8.2	8.2	8.19	8.25	8.19	8.19	8.34	8.12	8.29	8.25	8.12	8.19	8.3	8.25
Days	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
pH value	8.2	8.2	8.2	8.15	8.25	8.29	8.3	8.3	8.28	8.27	8.19	8.28	8.3	8.4	8.3
Days	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
pH value	8.11	8.35	8.22	8.24	8.29	8.3	8.15	8.14	8.24	8.11	8.13	8.11	8.15	8.14	8.2
Days	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
pH value	8.36	8.26	8.11	8.32	8.11	8.33	8.42	8.32	8.41	8.39	8.37	8.31	8.29	8.26	8.27

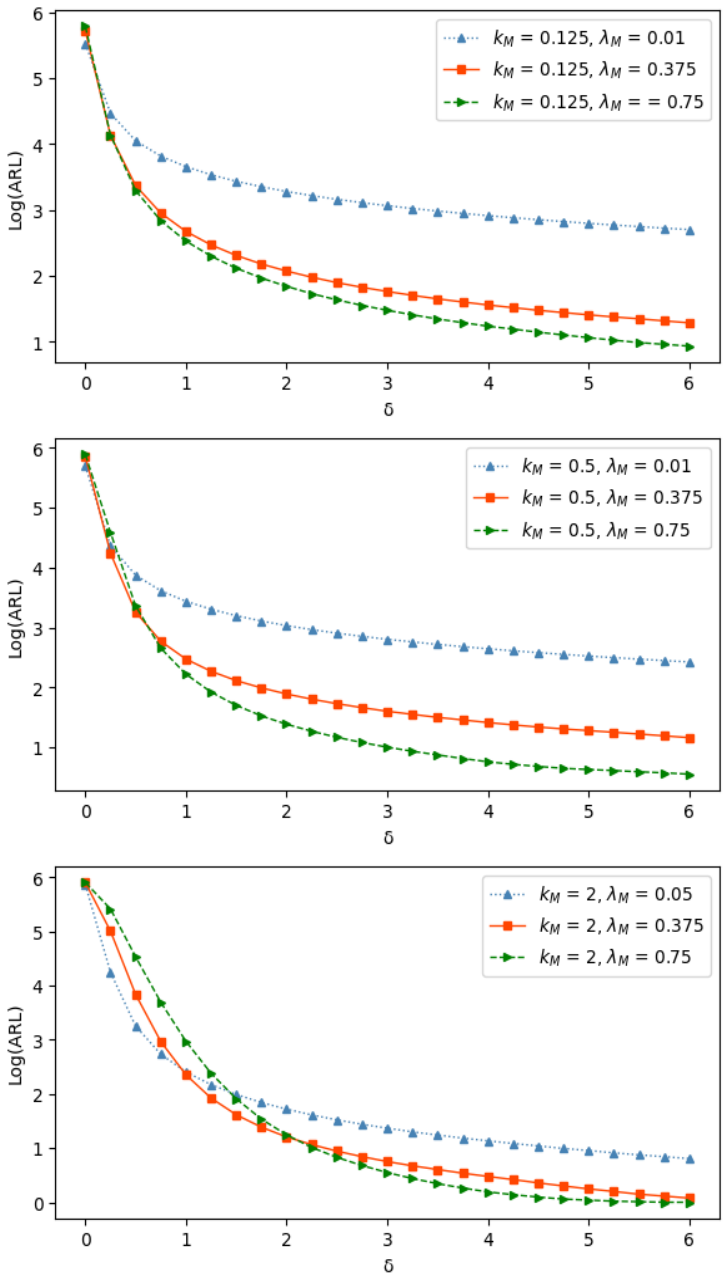


Figure 1: Steady-state ARL profiles (in logarithm scale) of nine designs of MEC control chart for shifts $\delta = 0 \; to \; \delta = 6$. Zero-state $ARL_0 \cong 370$.

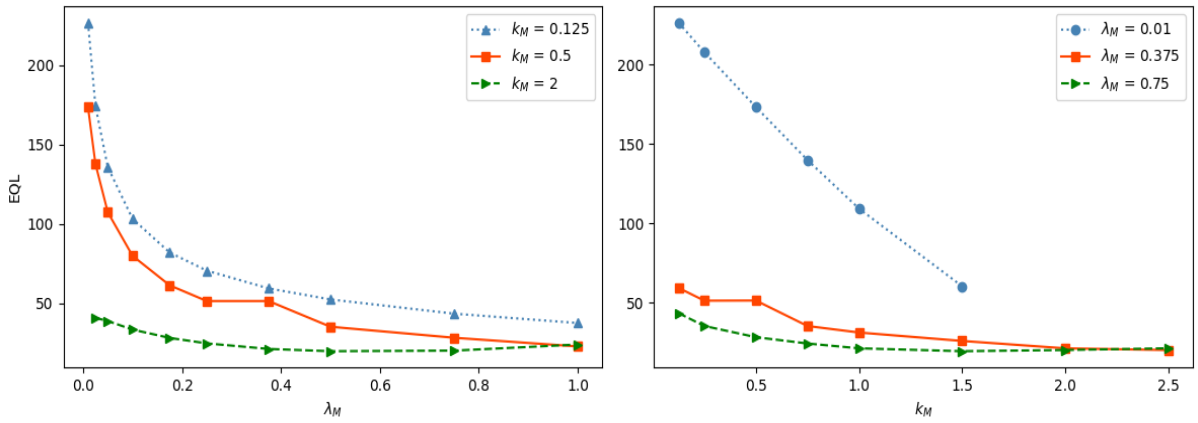


Figure 2: Effect of k_M and $\lambda_M$on EQL performance of MEC chart.

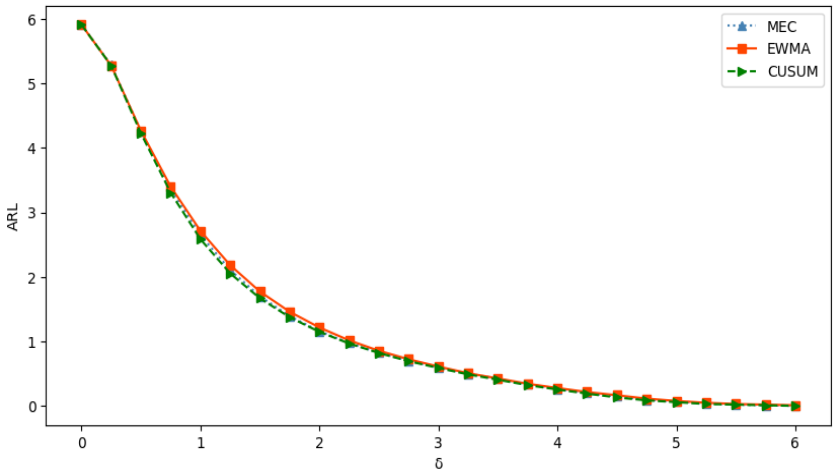


Figure 3: Steady-state ARL profiles (in logarithm scale) of MEC, EWMA, and CUSUM charts having minimum EQL values.

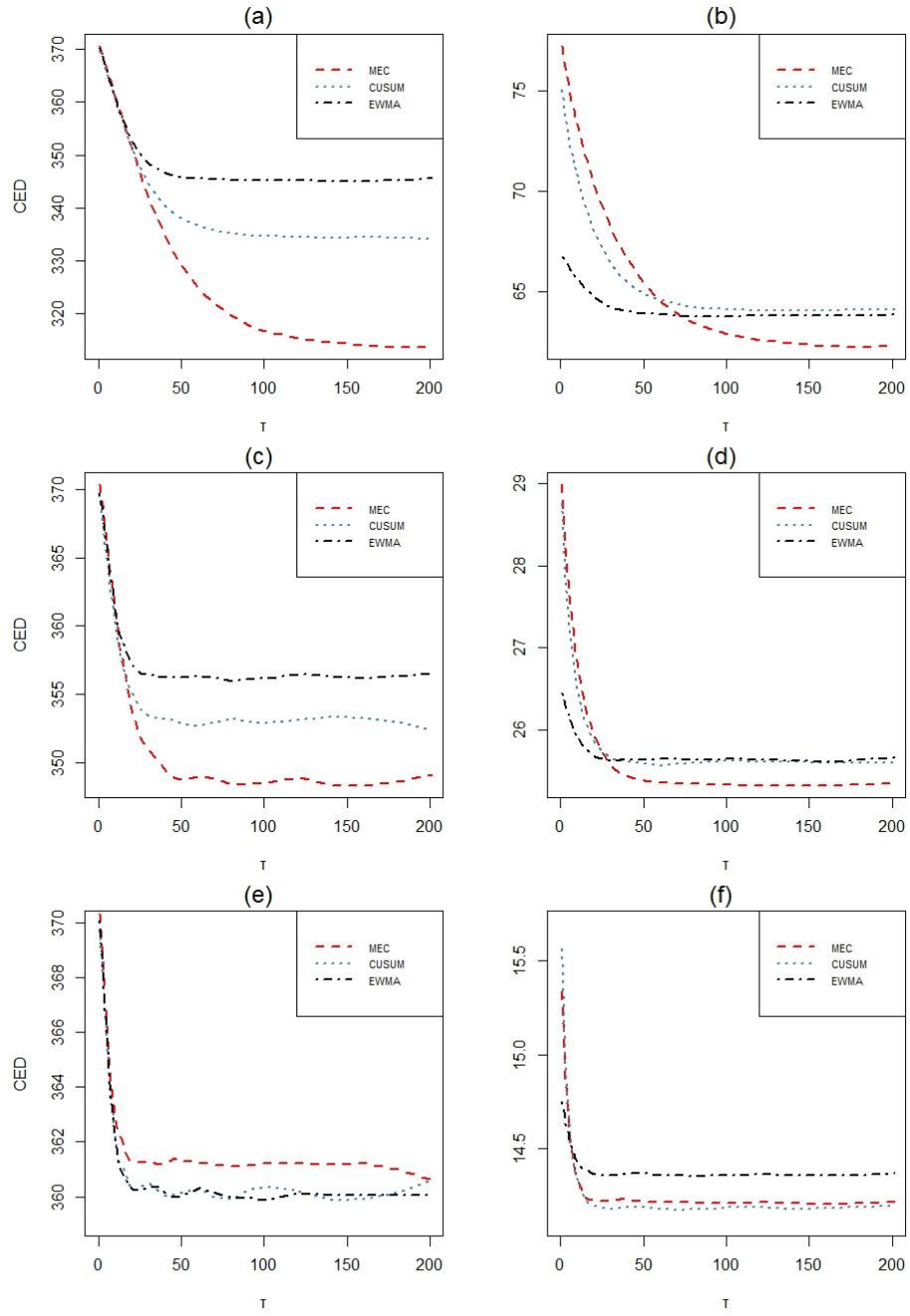


Figure 4: CED Performance Comparisons for EWMA, CUSUM, and MEC Charts Under IC (a, c, e) and OOC (b, d, f) Conditions at Different Shift Sizes, $\delta = 0.25$ (a and b), $\delta = 0.5$ (c and d), $\delta = 0.75$ (e and f). Zero-state $ARL_0 \cong 370$.

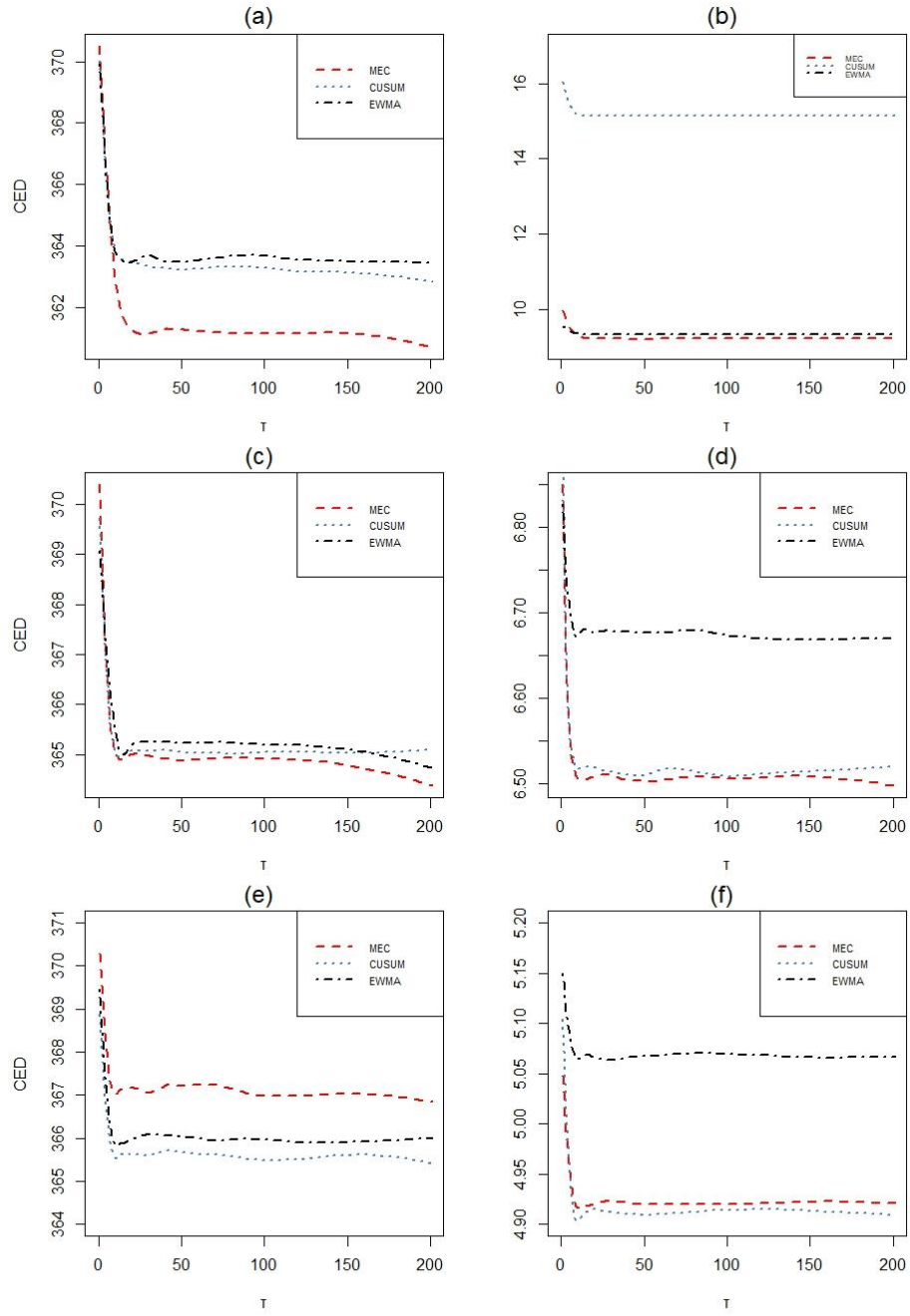


Figure 5: CED Performance Comparisons for EWMA, CUSUM, and MEC Charts Under IC (a, c, e) and OOC (b, d, f) Conditions at Different Shift Sizes, $\delta = 1$ (a and b), $\delta = 1.25$ (c and d), $\delta = 1.5$ (e and f). Zero-state $ARL_0 \cong 370$.

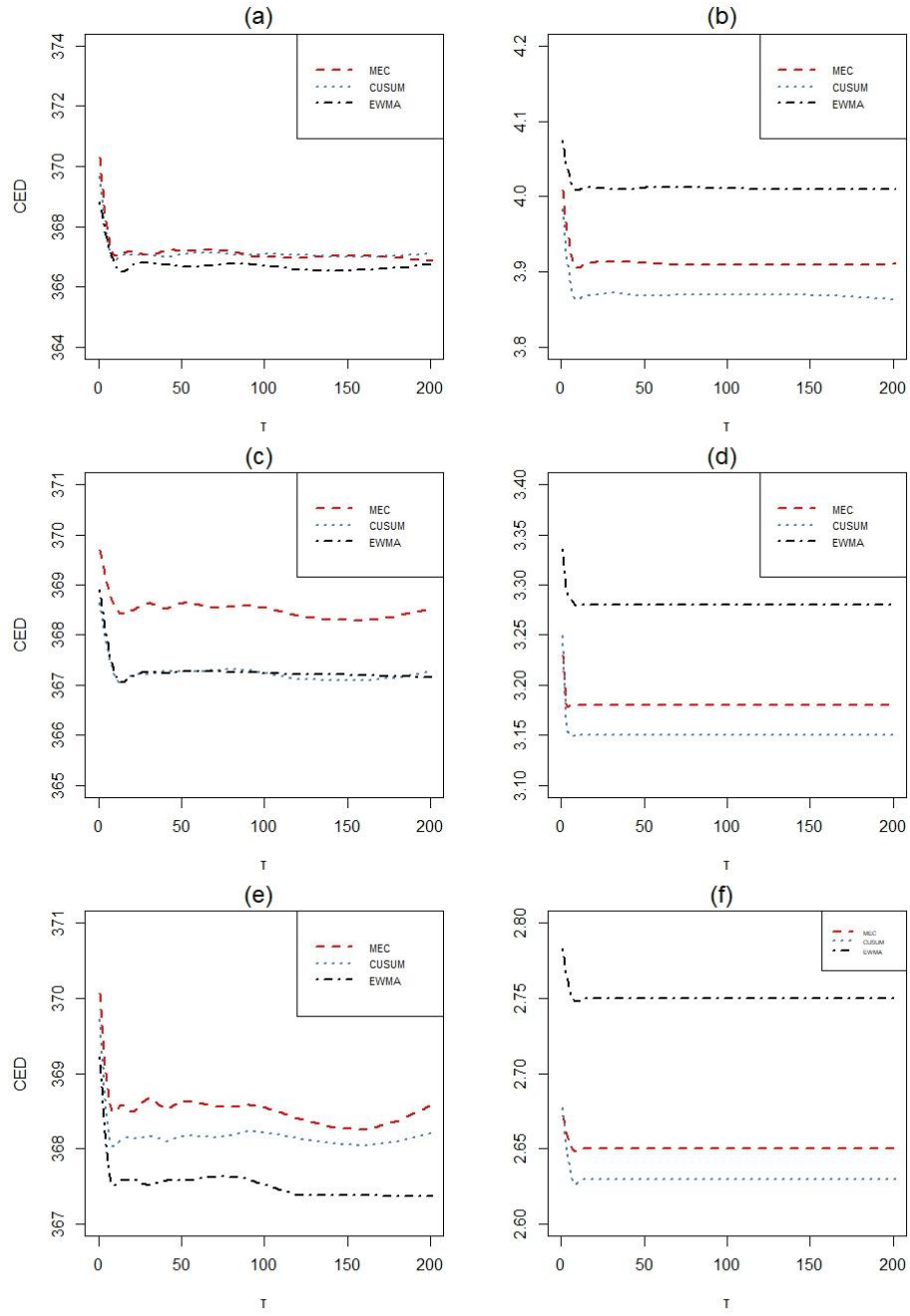


Figure 6: CED Performance Comparisons for EWMA, CUSUM, and MEC Charts Under IC (a, c, e) and OOC (b, d, f) Conditions at Different Shift Sizes, $\delta = 1.75$ (a and b), $\delta = 2$ (c and d), $\delta = 2.25$ (e and f). Zero-state $ARL_0 \cong 370$.

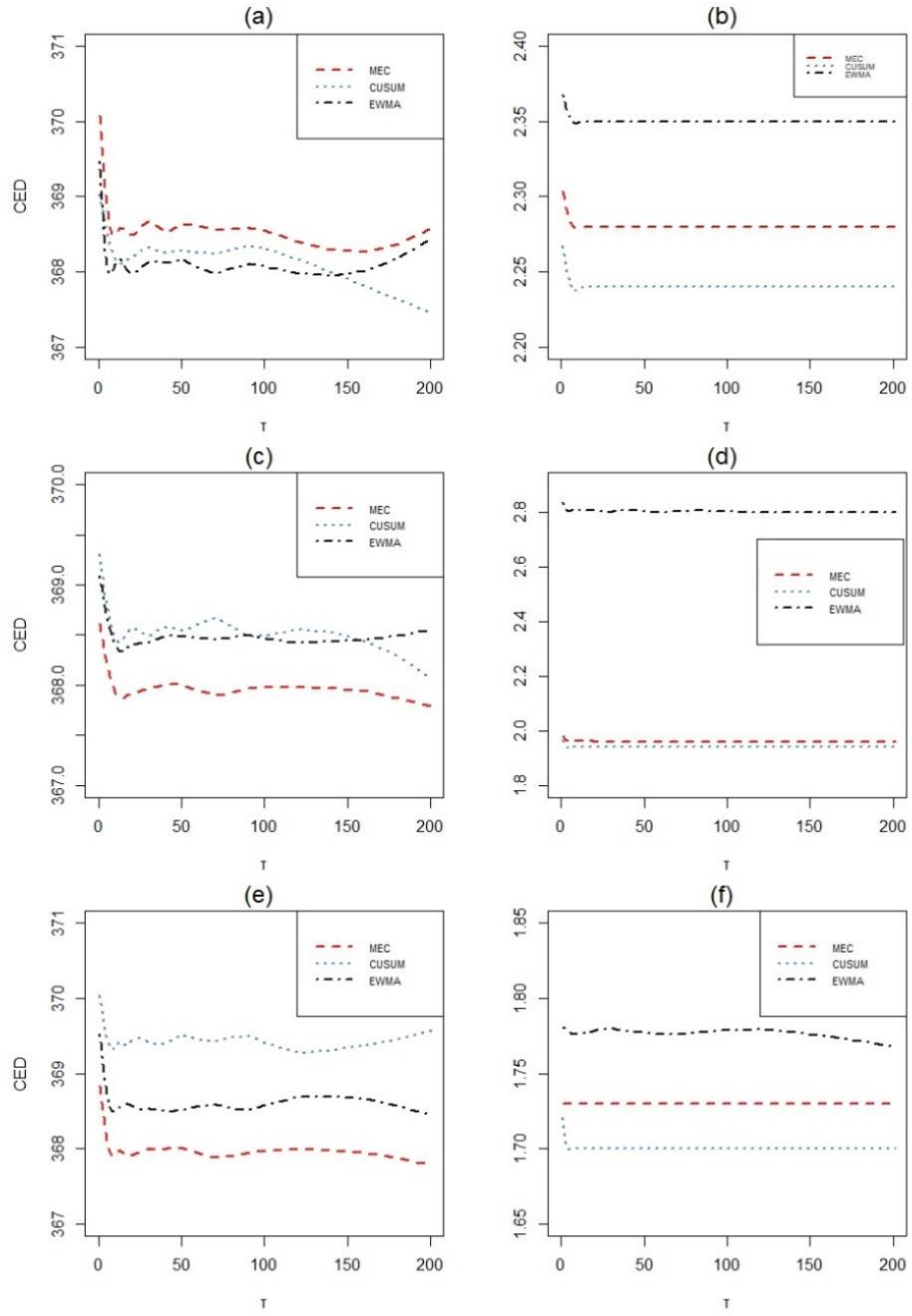


Figure 7: CED Performance Comparisons for EWMA, CUSUM, and MEC Charts Under IC (a, c, e) and OOC (b, d, f) Conditions at Different Shift Sizes, $\delta = 2.5$ (a and b), $\delta = 2.75$ (c and d), $\delta = 3$ (e and f). Zero-state $ARL_0 \cong 370$.

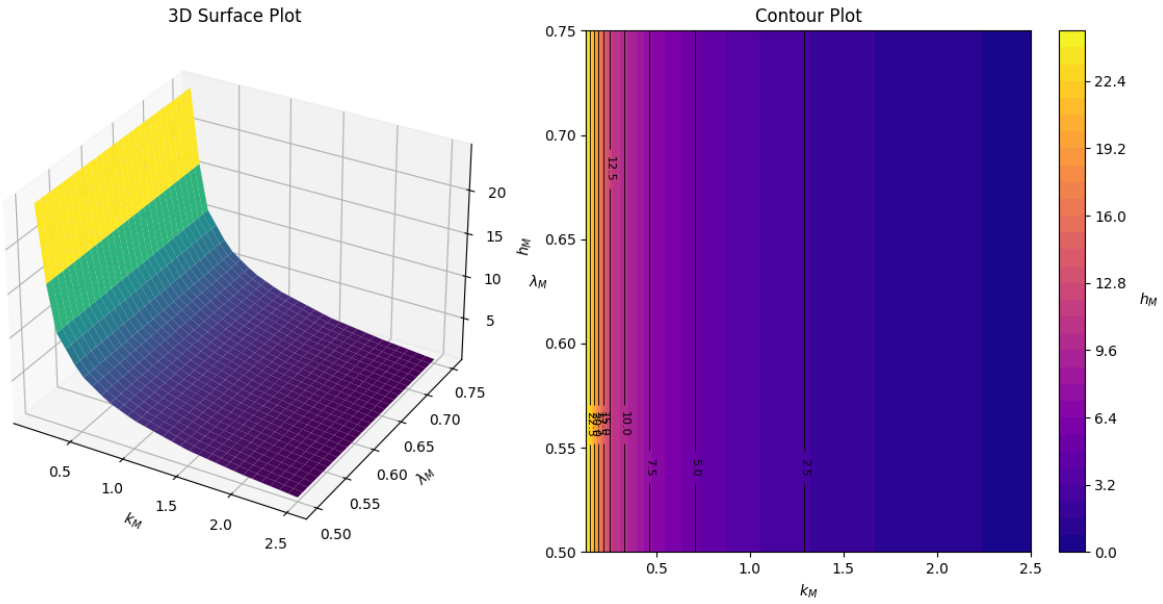


Figure 8: 3D Surface Plot and Contour Plot of $k_M, \lambda_M,$ and h_M

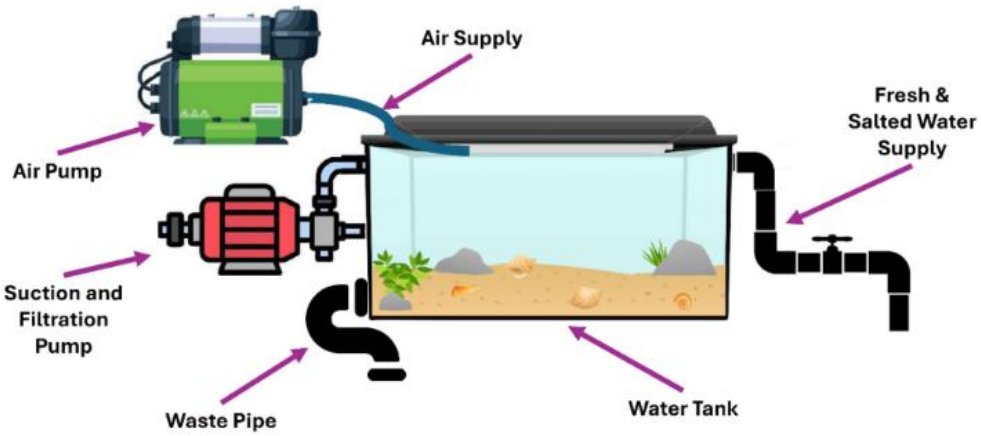


Figure 9: Mysid Culture Process Illustration [29]

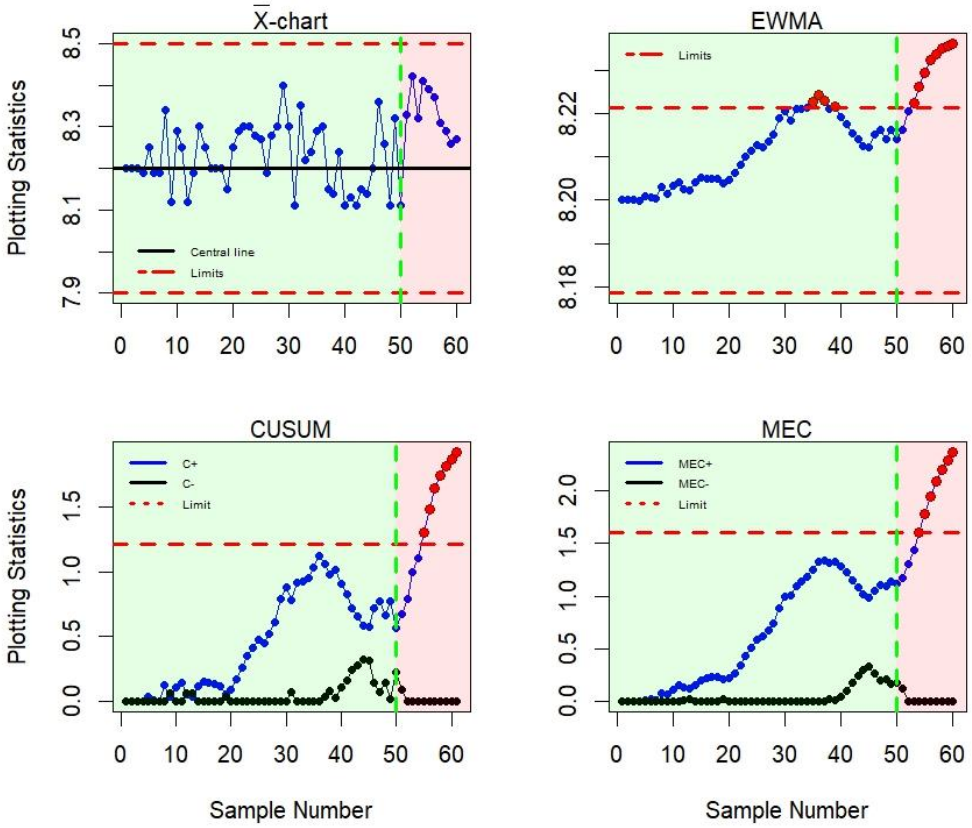


Figure 10. Application of Shewhart, EWMA, CUSUM, and MEC control charts on the pH data.

References

[1] J. Peluso, C. S. Pérez Coll, D. Cristos, D. E. Rojas, and C. M. Aronzon, "Comprehensive assessment of water quality through different approaches: Physicochemical and ecotoxicological parameters," *Science of The Total Environment,* vol. 800, p. 149510, 2021/12/15/ 2021, doi: https://doi.org/10.1016/j.scitotenv.2021.149510.

[2] L. S. Miranda, G. A. Ayoko, P. Egodawatta, and A. Goonetilleke, "Adsorption-desorption behavior of heavy metals in aquatic environments: Influence of sediment, water and metal ionic properties," *J. Hazard. Mater.,* vol. 421, p. 126743, 2022/01/05/ 2022, doi: https://doi.org/10.1016/j.jhazmat.2021.126743.

[3] A. S. Cardwell, P. H. Rodriguez, W. A. Stubblefield, D. K. DeForest, and W. J. Adams, "Chronic Toxicity of Iron to Aquatic Organisms under Variable pH, Hardness, and Dissolved Organic Carbon Conditions," *Environmental Toxicology and Chemistry,* vol. 42, no. 6, pp. 1371-1385, 2023/06/01 2023, doi: https://doi.org/10.1002/etc.5627.

[4] D. C. Montgomery, *Introduction to Statistical Quality Control, 8th Edition*. Hoboken, NJ, USA: Wiley (in English), 2019.

[5] H. Babamoradi, F. van den Berg, and Å. Rinnan, "Comparison of bootstrap and asymptotic confidence limits for control charts in batch MSPC strategies," *Chemometrics and Intelligent Laboratory Systems,* vol. 127, pp. 102-111, 2013, doi: https://doi.org/10.1016/j.chemolab.2013.06.005.

[6] A. B. A. Dawod, N. A. Adegoke, and S. A. Abbasi, "Efficient linear profile schemes for monitoring bivariate correlated processes with applications in the pharmaceutical industry," *Chemometrics Intellig. Lab. Syst.,* vol. 206, p. 104137, 2020/11/15/ 2020, doi: https://doi.org/10.1016/j.chemolab.2020.104137.

[7] H. Mitsutake, E. Guimarães, H. C. Freitas, L. C. Gontijo, D. Q. Santos, and W. Borges Neto, "Multivariate control charts based on NAS and mid-infrared spectroscopy for quality control of B5 blends of methyl soybean biodiesel in diesel," *Journal of Chemometrics,* vol. 29, no. 7, pp. 411-419, 2015/7// 2015, doi: https://doi.org/10.1002/cem.2720.

[8] W. A. Shewhart, "Economic Quality Control of Manufactured Product," *Bcl] Syst. Tech. J.,* vol. 9, no. 2, pp. 364-389, 1930/04/01 1930, doi: https://doi.org/10.1002/j.1538-7305.1930.tb00373.x.

[9] S. W. Roberts, "Control Chart Tests Based on Geometric Moving Averages," *Technometrics,* vol. 1, no. 3, pp. 239-250, 1959/08/01 1959, doi: https://doi.org/10.2307/1266443.

[10] E. S. Page, "Continuous inspection schemes," *Biometrika,* vol. 41, no. 1-2, pp. 100-115, 1954, doi: https://doi.org/10.2307/2333009.

[11] N. Abbas, M. Riaz, and R. J. M. M. Does, "Mixed Exponentially Weighted Moving Average–Cumulative Sum Charts for Process Monitoring," *Qual. Reliab. Eng. Int.,* https://doi.org/10.1002/qre.1385 vol. 29, no. 3, pp. 345-356, 2013/04/01 2013, doi: https://doi.org/10.1002/qre.1385.

[12] J. O. Ajadi, M. Riaz, and K. Al-Ghamdi, "On increasing the sensitivity of mixed EWMA–CUSUM control charts for location parameter," *Journal of Applied Statistics,* vol. 43, no. 7, pp. 1262-1278, 2016/5// 2016, doi: 10.1080/02664763.2015.1094453.

[13] H. Z. Nazir, N. Abbas, M. Riaz, and R. J. M. M. Does, "A comparative study of memory-type control charts under normal and contaminated normal environments," *Quality and Reliability Engineering International,* vol. 32, no. 4, pp. 1347-1356, 2016/6// 2016, doi: 10.1002/qre.1835.

[14] J. O. Ajadi and M. Riaz, "Mixed multivariate EWMA-CUSUM control charts for an improved process monitoring," *Communications in Statistics - Theory and Methods,* vol. 46, no. 14, pp. 6980-6993, 2017/7// 2017, doi: 10.1080/03610926.2016.1139132.

[15] S. Hussain, X. Wang, S. Ahmad, and M. Riaz, "On a class of mixed EWMA-CUSUM median control charts for process monitoring," *Quality and Reliability Engineering International,* vol. 36, no. 3, pp. 910-946, 2020/4// 2020, doi: 10.1002/qre.2608.
[16] G. S. Rao, M. Aslam, U. Rasheed, and C.-H. Jun, "Mixed EWMA–CUSUM chart for COM-Poisson distribution," *Journal of Statistics and Management Systems,* vol. 23, no. 3, pp. 511-527, 2020/4// 2020, doi: 10.1080/09720510.2019.1639947.
[17] S. M. Anwar, M. Aslam, B. Zaman, and M. Riaz, "Mixed memory control chart based on auxiliary information for simultaneously monitoring of process parameters: An application in glass field," *Computers & Industrial Engineering,* vol. 156, p. 107284, 2021/06/01/ 2021, doi: 10.1016/j.cie.2021.107284.
[18] M. Shamsuzzaman, S. Haridy, A. Maged, H. Bashir, A. Shamsuzzoha, and A. Ali, "An effective statistical process control scheme for industrial environmental monitoring," *Chemometrics Intellig. Lab. Syst.,* vol. 229, p. 104651, 2022/10/15/ 2022, doi: https://doi.org/10.1016/j.chemolab.2022.104651.
[19] K. Talordphop, S. Sukparungsee, and Y. Areepong, "On designing new mixed modified exponentially weighted moving average - exponentially weighted moving average control chart," *Results in Engineering,* vol. 18, p. 101152, 2023/06/01/ 2023, doi: https://doi.org/10.1016/j.rineng.2023.101152.
[20] V. Alevizakos, K. Chatterjee, and C. Koukouvinos, "On the performance and comparison of various memory-type control charts," *Communications in Statistics - Simulation and Computation,* pp. 1-21, 2024, doi: https://doi.org/10.1080/03610918.2024.2310692.
[21] B. Zaman, M. Riaz, N. Abbas, and R. J. M. M. Does, "Mixed Cumulative Sum–Exponentially Weighted Moving Average Control Charts: An Efficient Way of Monitoring Process Location," *Qual. Reliab. Eng. Int.,* vol. 31, no. 8, pp. 1407-1421, 2015/12/01 2015, doi: https://doi.org/10.1002/qre.1678.
[22] B. Zaman, N. Abbas, M. Riaz, and M. H. Lee, "Mixed CUSUM-EWMA chart for monitoring process dispersion," *The International Journal of Advanced Manufacturing Technology,* vol. 86, no. 9, pp. 3025-3039, 2016/10/01 2016, doi: https://doi.org/10.1007/s00170-016-8411-0.
[23] M. Mansouri, M. F. Harkat, M. Nounou, and H. Nounou, "Midpoint-radii principal component analysis -based EWMA and application to air quality monitoring network," *Chemometrics Intellig. Lab. Syst.,* vol. 175, pp. 55-64, 2018/04/15/ 2018, doi: https://doi.org/10.1016/j.chemolab.2018.01.016.
[24] S. V. Crowder, "Design of exponentially weighted moving average schemes," *Journal of Quality technology,* vol. 21, no. 3, pp. 155-162, 1989.
[25] E. S. Page, "Control Charts for the Mean of a Normal Population," *Journal of the Royal Statistical Society: Series B (Methodological),* vol. 16, no. 1, pp. 131-135, 1954/01/01 1954, doi: https://doi.org/10.1111/j.2517-6161.1954.tb00154.x.
[26] X. Qin, J. Zhang, A. Mukherjee, D. Yu, and L. Liu, "Comparisons of some new and existing EWMA schemes for binomial count processes with applications to monitoring hospital No-Shows," *Quality Technology & Quantitative Management,* pp. 1-23, 2024, doi: https://doi.org/10.1080/16843703.2024.2411120.
[27] Y. Tan, A. Mukherjee, and J. Zhang, "Two Bayesian approaches of monitoring mean of Gaussian process using Bayes factor," *Qual. Reliab. Eng. Int.,* vol. 40, no. 6, pp. 3320-3339, 2024/10/01 2024, doi: https://doi.org/10.1002/qre.3567.
[28] L. K. Chan and J. Zhang, "Some issues in the design of EWMA Charts," *Communications in Statistics - Simulation and Computation,* vol. 29, no. 1, pp. 201-217, 2000/01/01 2000, doi: https://doi.org/10.1080/03610910008813610.
[29] F. Marini, "Culture Of Mysid Shrimp And Bivalve Trochphores (veligers)," (in English), *Reefs,* 2003, doi: https://reefs.com/magazine/the-breeder-s-net-culture-of-mysid-shrimp-and-bivalve-trochphores-veligers.

[30] G. Marsaglia and J. Marsaglia, "Evaluating the Anderson-Darling Distribution," *Journal of Statistical Software,* vol. 9, no. 2, pp. 1 - 5, 02/25 2004, doi: https://doi.org/10.18637/jss.v009.i02.
[31] M. Verbeek, *A Guide to Modern Econometrics, 5th Edition*. Hoboken, NJ, USA: Wiley (in English), 2017.